Evaluating Very Long-Term Conversational Memory of LLM Agents

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Abstract

Existing works on long-term open-domain dialogues focus on evaluating model responses within contexts spanning no more than five chat sessions. Despite advancements in long-context large language models (LLMs) and retrieval augmented generation (RAG) techniques, their efficacy in *very* long-term dialogues remains unexplored. To address this research gap, we introduce a machine-human pipeline to generate high-quality, *very* long-term dialogues by leveraging LLM-based agent architectures and grounding their dialogues on personas and temporal event graphs. Moreover, we equip each agent with the capability of sharing and reacting to images. The generated conversations are verified and edited by human annotators for long-range consistency and grounding to the event graphs. Using this pipeline, we collect LoCoMo, a dataset of *very* long-term conversations, each encompassing approx. 600 turns and 16K tokens on avg., over up to 32 sessions. Based on LoCoMo, we present a comprehensive evaluation benchmark to measure long-term memory in models, encompassing question answering, event summarization, and multi-modal dialogue generation tasks. Our experimental results indicate that LLMs exhibit challenges in understanding lengthy conversations and comprehending long-range temporal and causal dynamics within dialogues. Employing strategies like long-context LLMs or RAG can offer improvements but these models still substantially lag behind human performance.[1]

1 Introduction

Despite recent advancements in dialogue models based on LLMs for extended contexts (Bertsch et al., 2023; Xiao et al., 2023), as well as the integration of retrieval augmented generation (RAG)

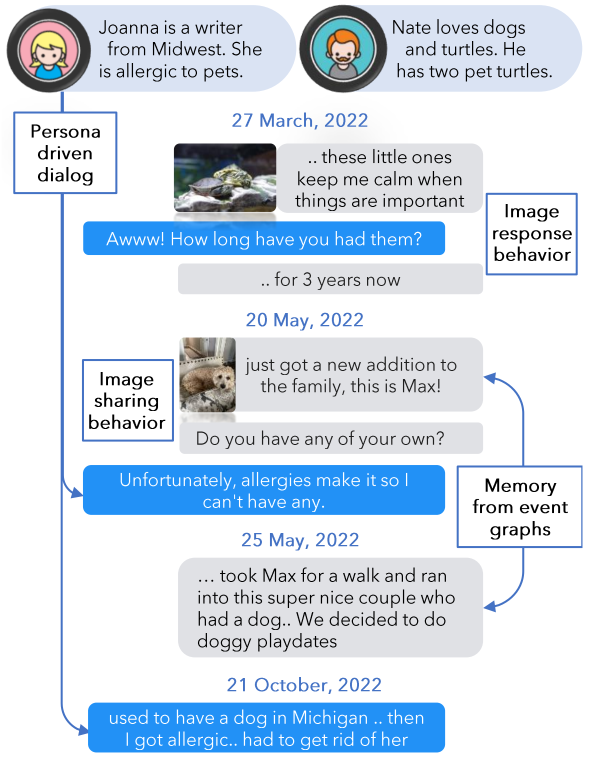

Figure 1: **An example in LoCoMo.** Dialogs are steered by the speakers' personas and corresponding events e.g., Joanna's responses are consistent with her pet allergies. For Nate, the event *got a new dog* is followed by a *playdate with neighbor's dog*, showcasing long-term memory. Multimodal dialog is enabled with image-sharing and image-response behaviors.

techniques (Shuster et al., 2021; Ram et al., 2023; Shi et al., 2023), there is still a need for thorough evaluation of their efficacy in handling very long conversations. Indeed, studies in long-term open-domain dialogues have concentrated on assessing model responses within limited contexts e.g., ~1K tokens over five chat sessions (Xu et al., 2022; Jang et al., 2023b; Zhang et al., 2023). This long term evaluation is crucial for refining engaging chatbots capable of remembering key information from past interactions, to generate empathetic, consistent, and useful responses.

[1]Code and data are at https://snap-research.github.io/locomo

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评估大语言模型代理的长期对话记忆

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摘要

现有的长期开放域对话研究主要关注在不超过五个聊天会话的上下文中评估模型响应。尽管长上下文大语言模型（LLM）和检索增强生成（RAG）技术取得了进展，但它们在<样式 id='1'>非常</样式>长期的对话中的有效性仍未被探索。为了解决这一研究空白，我们引入了一个机器-人工管道，通过利用基于LLM的代理架构，并使其对话基于角色和时间事件图来生成高质量的<样式 id='3'>非常</样式>长期对话。此外，我们还为每个代理配备了共享和反应图像的能力。生成的对话由人工标注员验证和编辑，以确保长程一致性和与事件图的关联性。使用该管道，我们收集了LOCOMO，一个包含<样式 id='5'>非常</样式>长期对话的数据集，每个对话包含约600轮和平均16K个token，跨越多达32个会话。基于LOCOMO，我们提出了一套全面的评估基准，用于衡量模型中的长期记忆，包括问答、事件摘要和多模态对话生成任务。我们的实验结果表明，LLM在理解长对话和理解对话中的长程时间和因果关系方面存在挑战。采用长上下文LLM或RAG等策略可以提供改进，但这些模型仍远落后于人类表现[1]

1 引言

尽管基于大语言模型（LLMs）的对话模型在扩展上下文方面取得了近期进展（Bertsch等人,2023；Xiao等人， 2023），以及检索增强生成（RAG）的集成

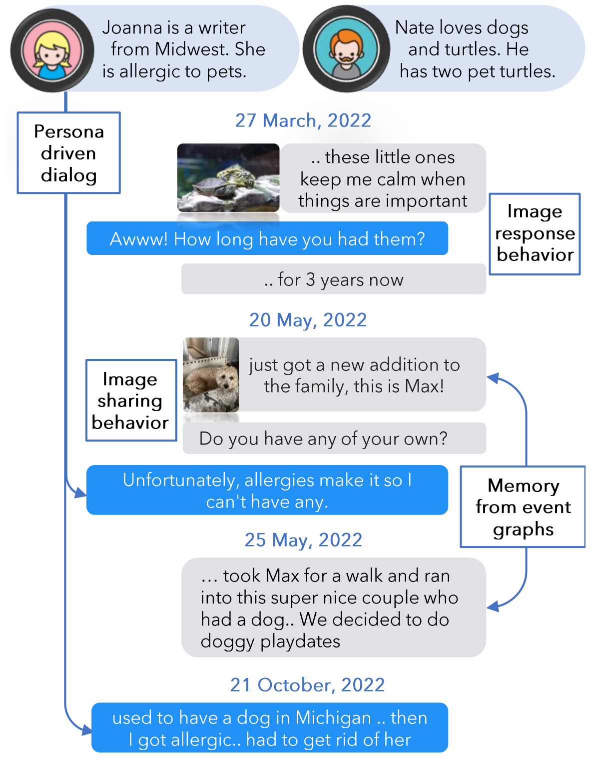

图1：**LOCOMO的一个示例oCMOo。** 对话由说话人的角色和相应的事件引导，例如，乔安娜的回应与其宠物过敏一致。对于内特，事件得到一只新狗后跟着是与邻居的狗一起玩耍，展示了长期记忆。通过图像共享和图像响应行为实现了多模态对话。

技术（Shuster等人，2021； Ram等人，2023；Shi等人，2023），在处理超长期对话方面，他们的有效性仍需全面评估。事实上，长期开放域对话的研究主要集中在评估模型在有限语境内的响应，例如 ~1K tokens跨越五个对话会话（Xu等人，2022；张等人，2023b；张等人，2023）。这种长期评估对于改进能够记住过去交互关键信息的聊天机器人至关重要，从而生成富有同情心、一致且有用的响应。

[1]代码和数据位于 https://snap-research.github.io/locomo

[†]平等建议。

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Dataset	Avg. turns per conv.	Avg. sessions per conv.	Avg. tokens per conv.	Time Interval	Multimodal	Collection
MPChat (Ahn et al., 2023)	2.8	1	53.3	-	✓	Reddit
MMDialog (Feng et al., 2022)	4.6	1	72.5	-	✓	Social media
Daily Dialog (Li et al., 2017)	7.9	1	114.7	-	✗	Crowdsourcing
SODA (Kim et al., 2023)	7.6	1	122.4	-	✗	LLM-generated
MSC (Xu et al., 2022) (train; 1-4 sessions)	53.3	4	1,225.9	few days	✗	Crowdsourcing
Conversation Chronicles (Jang et al., 2023a)	58.5	5	1,054.7	few hours - years	✗	LLM-generated
LoCoMo (ours)	588.2	27.2	16,618.1	few months	✓	LLM-gen. + crowdsourc.

Table 1: **Statistics of LoCoMo** compared to existing dialog datasets. The average length of a conversation in LoCoMo is 16x that of MSC (Xu et al., 2022), distributed over 10x more turns and 5x more sessions (on average).

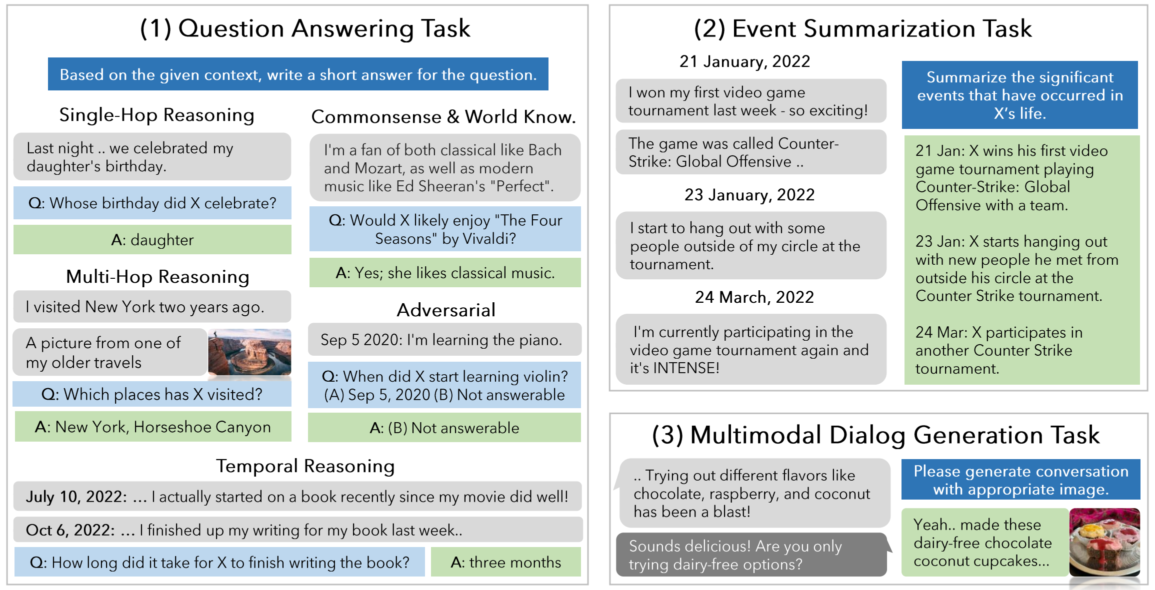

Figure 2: **Overview of our evaluation framework**. We propose three tasks: question answering, event summarization and multimodal dialog generation to evaluate models' comprehension in very long-term dialogues.

To this end, we present the first study of very long-term open-domain multi-modal dialogues, closely mirroring real-world online interactions, collected via a human-machine pipeline where we first use LLM-based generative agents to generate conversations and then ask human annotators to fix any long-term inconsistencies in the conversations. Specifically, drawing on the understanding that real-world conversations are a complex blend of collective memories (Assmann and Czaplicka, 1995; Hirst and Manier, 2008), individual viewpoints (Hirst et al., 2018), external influences (Hirst and Echterhoff, 2012), and the unique persona of the speakers (Pruitt and Grudin, 2003; Cooper, 1999; Zhou et al., 2020; Shum et al., 2019), we create *very long-term* dialogues based on LLM agent with the following features: (1) a unique persona (§3.1); (2) a timeline of causally interlinked events in their lives (§3.2); and (3) *reflect & response* mechanism to respond based on dialogue history (like in Park et al. (2023)) and *image sharing & image reaction* behavior which sends or reacts to images (§3.3). Finally, human annotators fix long-range inconsistencies in dialogues, remove irrelevant images, and verify the grounding of dialogs to events (§3.4). With this pipeline, we create LoCoMo, a dataset of 10 *very long-term* dialogues, each consisting of 600 turns and 16K tokens on avg., over up to 32 sessions (see Figure 1, Table 1).

Conventional approaches for evaluating conversational agents in open-domain dialogues involves directly evaluating the agent response based on past dialogue history. It often employs lexical overlap (Papineni et al., 2002) and semantic overlap (Zhang et al., 2019) between ground truth and the agent response, or consistency (Ghazarian et al., 2022), contradiction (Nie et al., 2021; Welleck et al., 2019), and empathy (Zhang et al., 2021a, 2022) of the agent response. However, these evaluation metrics are not well-suited for directly assessing the agent's comprehension of long-term contexts. In this study, we present a holistic evaluation framework to assess an agent's proficiency in managing and responding within long-term contexts (see Figure 2). First, agents need to "recall" past context correctly to integrate relevant infor-

数据集	平均回合数 每对话	平均会话数 每对话	平均令牌数 每对话	时间间隔	多模态	集合
MPChat(Ahn等人, 2023)	2.8	1	53.3	-	✓	Reddit
MMDialog(冯等人, 2022)	4.6	1	72.5	-	✓	社交媒体
每日对话 (李等人, 2017)	7.9	1	114.7	-	✗	众包
SODA (金等人,2023)	7.6	1	122.4	-	✗	大语言模型生成
MSC (徐等人, 2022) (训练；1-4次会话)	53.3	4	1,225.9	几天	✗	众包
对话编年史 (张等人,2023a)	58.5	5	1,054.7	几小时-几年	✗	大语言模型生成
LOCOMO (我们的)	588.2	27.2	16,618.1	几个月	✓	LLM生成 +众包

表1：**LCOOMO 与现有对话数据集的统计对比。LCOOMO 中对话的平均长度是 MSC 的 16 倍**（徐等人, 2022），分布在 10 倍更多的回合和 5 倍更多的会话中（平均而言）。

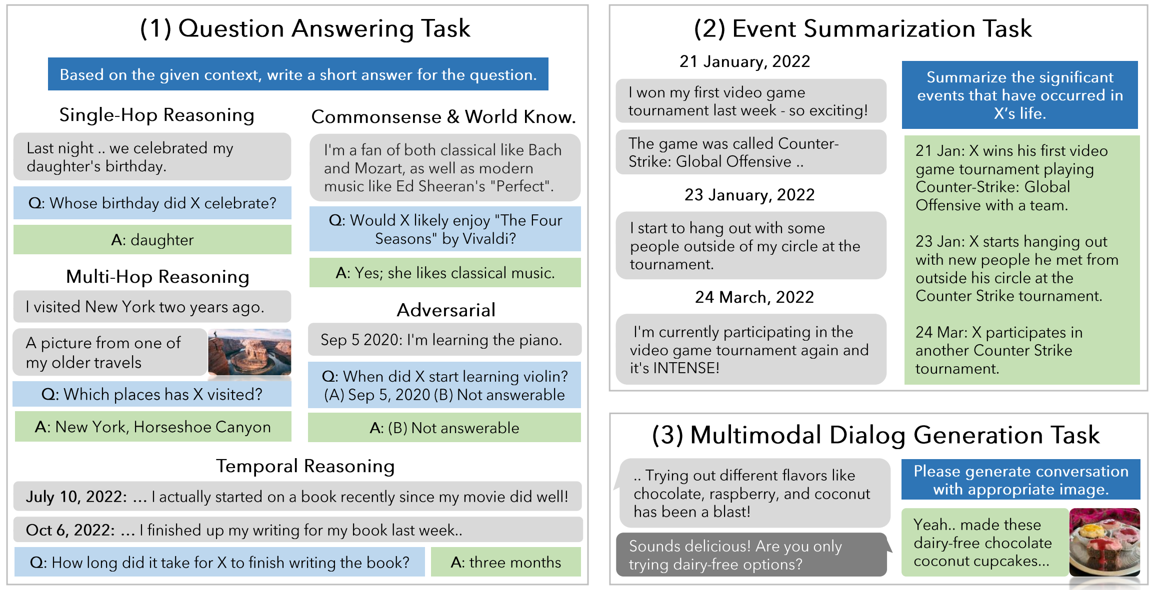

图2：**我们评估框架的概述**。我们提出了三个任务：问答、事件摘要和多模态对话生成，以评估模型在非常长期的对话中的理解能力。

为此，我们首次研究了超长期开放域多模态对话，其紧密模拟真实世界的在线交互，通过人机流程收集，我们首先使用基于LLM的生成式代理生成对话，然后要求人类标注员修正对话中的长期不一致性。具体而言，基于对现实世界对话的理解——它们是集体记忆（Assmann和Czaplicka，1995；Hirst和Manier，2008）、个人观点（Hirst等人，2018）、外部影响（Hirst和Echterhoff，2012）以及说话人独特人格（Pruitt和Grudin，2003；Cooper，1999；Zhou等人，2020；Shum等人，2019）的复杂混合（§3.1），我们基于LLM代理创建了具有以下特征的超长期对话：（1）独特人格（§3.1）；（2）其生活中因果关联事件的时间线（§3.2）；以及（3）反映与响应机制，根据对话历史进行响应（如Park等人（2023）所述）和图像分享与图像反应行为，发送或反应图像（§3.3）。最后，人类标注员修正长期在对话中处理范围不一致，移除无关图像，并验证对话到事件的 grounding（§3.4）。通过这个流程，我们创建了 LCOOMO，一个包含 10 个非常长期的对话的数据集，每个对话包含 600 个回合和平均 16K 个 token，跨越最多 32 个会话（见图1，表1）。

传统方法用于评估开放域对话中的对话代理，通常基于过去的对话历史直接评估代理的响应。它经常采用真实文本与代理响应之间的词汇重叠（Papineni 等人， 2002）和语义重叠（Zhang 等人，2019），或一致性（Ghazarian 等人，2022），矛盾（Nie 等人，2021； Welleck 等人，2019），以及代理响应的共情（Zhang 等人，2021a，2022）。然而，这些评估指标并不适合直接评估代理对长期上下文的理解能力。在本研究中，我们提出一个综合评估框架，用于评估代理在长期上下文中管理和响应的能力（见图 2）。首先，代理需要正确"召回"过去的上下文，以便将相关信息整合到未来的响应中。我们通过问答（QA）任务（§4.1）直接考察他们的 记忆，(QA) 任务（§4.1）。我们将问题分为五种不同的推理类型，从多个角度评估记忆：单跳、多跳、时间、常识或世界知识，以及对抗性。其次，代理还需要识别对话中的长期因果和时间联系，以生成共情和相关的响应。我们提出一个事件图摘要任务（§4.2）来衡量他们的因果和时间理解能力。在这个任务中，与每个 LLM 说话人相关的事件图作为正确答案，模型需要从对话历史中提取这些信息。第三，对话代理需要利用从过去对话中"召回"的相关上下文，生成与当前叙事一致的响应。我们通过多模态对话生成任务（§4.3）评估这种能力。

mation into future responses. We present a direct examination of their *memory* via a *question answering* (QA) task (§4.1). We classify questions into five distinct reasoning types to evaluate memory from multiple perspectives: single-hop, multi-hop, temporal, commonsense or world knowledge, and adversarial. Second, agents also need to recognize long-range causal and temporal connections in the dialogues to generate empathetic and relevant responses. We propose a measurement of their causal and temporal understanding with an *event graph summarization* task (§4.2). In this task, the event graphs linked to each LLM speaker serve as the correct answers, and models are tasked with extracting this information from the conversation history. Third, conversational agents need to utilize relevant context recalled from past conversations to generate responses that are consistent with the ongoing narrative. We assess this ability via the *multi-modal dialog generation* task (§4.3).

We present extensive experimental results on the LoCoMo benchmark using instruction-based LLMs, long-context LLMs, and RAG techniques (§5). Our findings include: (1) Long-context LLMs and RAG demonstrate effectiveness in QA tasks, improving 'memory' capabilities of LLMs (with improvements ranging from 12-20%), but still significantly lag behind human levels (by 36%), especially in temporal reasoning, (by 41%); (2) long-context LLMs demonstrate significant difficulty with adversarial questions in the QA task, showing a performance that is 65% lower than the base model. They are especially prone to misassigning dialogs or events to the wrong speaker. Moreover, they show poor performance on *event graph summarization*, indicating that they may grasp the factual elements within the entire conversation but do not accurately comprehend the context; and (3) RAG offers a balanced compromise, combining the accuracy of short-context LLMs with the extensive comprehension of wide-context LLMs, and does particularly well when dialogues are transformed into a database of assertions (*observations*) about each speaker's life and persona.

2 Related Work

Long-term Dialogue. Recent approaches involve retrieving historical context from a range of previous dialogues and reasoning over retrieved segments in a temporal order (Lee et al., 2023b; Lu et al., 2023; Zhong et al., 2023; Liang et al., 2023) and/or using events to scaffold the dialogues (Jang et al., 2023b; Zhang et al., 2023) to enable consistency in long-term conversations. Some limitations of such frameworks are: (1) The accuracy of retrieval can be compromised, as the retrieval model is generally trained on tasks focusing on semantic similarity rather than specifically on such dialogues. Additionally, real-world dialogues often feature co-references and missing content (*i.e.,* anaphora) (Anantha et al., 2021), which further complicate the retrieval process (Mallen et al., 2023; Gao et al., 2023b; Liu et al., 2023b); (2) Challenges arise in reasoning over retrieved documents, especially when the model struggles to identify the correct context among the retrieved data (Liu et al., 2023a); (3) Reasoning over time intervals presents challenges. For example, the way a system responds about past events can vary depending on the amount of time that has passed since the last conversation (Zhang et al., 2023; Jang et al., 2023b). Therefore, it is essential to have conversations of considerable length, as well as a systematic evaluation framework, to accurately assess the effectiveness of approaches to long-term dialogue generation. We design a long-term conversation generation pipeline based on retrieval augmentation and events graphs and propose a framework for evaluating long-term dialog agents.

Multi-modal Dialogue. Multi-modal dialogue primarily consists of two types of tasks: image-grounded dialogue and image-sharing dialogue. The image-grounded dialogue task is centered around responding to questions (Antol et al., 2015; Das et al., 2017; Kottur et al., 2019) or creating natural conversations related to specific images (Mostafazadeh et al., 2017; Shuster et al., 2018; Meng et al., 2020; Zheng et al., 2021). Conversely, the image-sharing dialogue task focuses on selecting images that semantically align with the provided dialogue context (Zang et al., 2021; Feng et al., 2022; Lee et al., 2023c). We use a method from the image-sharing dialogue task to create multimodal dialogs which are then evaluated as an image-grounded dialogue task.

Synthetic Evaluation Benchmark. Faced with a shortage of human-generated data and observing that LLMs are approaching the quality of human-level annotations (He et al., 2023; Lee et al., 2023a), there has been a surge in research drawing inspiration from this development. Consequently, numerous studies have started utilizing LLMs to

首先，代理需要正确“召回”过去的上下文，以便将相关信息整合到未来的响应中。我们通过问答（QA）任务（§4.1）直接考察他们的记忆，(QA) 任务（§4.1）。我们将问题分为五种不同的推理类型，从多个角度评估记忆：单跳、多跳、时间、常识或世界知识，以及对抗性。其次，代理还需要识别对话中的长期因果和时间联系，以生成共情和相关的响应。我们提出一个事件图摘要任务（§4.2）来衡量他们的因果和时间理解能力。在这个任务中，与每个 LLM 说话人相关的事件图作为正确答案，模型需要从对话历史中提取这些信息。第三，对话代理需要利用从过去对话中“召回”的相关上下文，生成与当前叙事一致的响应。我们通过多模态对话生成任务（§4.3）评估这种能力。

我们展示了基于指令的大语言模型、长上下文大语言模型和RAG技术的OCOM基准测试的详尽实验结果（§5）。我们的发现包括：(1) 长上下文大语言模型和RAG在问答任务中表现出有效性，提升了大语言模型的‘记忆’能力（改进幅度在12-20%之间），但仍显著落后于人水平（落后36%），尤其是在时间推理方面（落后41%）；(2) 长上下文大语言模型在问答任务中的对抗性问题方面表现出显著困难，性能比基础模型低65%。它们尤其容易将对话或事件错误地分配给错误的说话人。此外，它们在事件图摘要方面表现不佳，表明它们可能掌握了整个对话中的事实元素，但无法准确理解上下文；以及(3) RAG提供了一个平衡的折中方案，结合了短上下文大语言模型的准确性以及宽上下文大语言模型的广泛理解能力，当对话被转换为关于每个说话人生活和角色的断言（观察）数据库时表现尤其出色。

2 相关工作

长期对话。 近期方法涉及从一系列先前对话中检索历史上下文，并按时间顺序对检索到的片段进行推理（Lee等人， 2023b；Lu等人， 2023；Zhong等人， 2023；Liang等人， 2023）和/或使用事件来构建对话（张等人， 2023b；张等人， 2023）以实现长期对话的一致性。此类框架的一些限制包括：(1) 检索的准确性可能受到影响，因为检索模型通常在关注语义相似性而非特定对话的任务上进行训练。此外，现实对话中常出现共指和缺失内容（即指代）（Anantha等人， 2021），这进一步增加了检索的复杂性（Mallen等人，2023；Gao等人， 2023b； Liu等人， 2023b）；(2) 在检索到的文档上进行推理时存在挑战，特别是当模型难以在检索数据中识别正确上下文时（Liu等人， 2023a）；(3) 对时间间隔进行推理存在挑战。例如，系统对过去事件的回应方式可能因距上次对话的时间长短而异（张等人，2023； 张等人，2023b）。因此，需要具有相当长度的对话，以及一个系统的评估框架，以准确评估长期对话生成方法的有效性。我们设计了一个基于检索增强和事件图的长期对话生成管道，并提出了一个评估长期对话代理的框架。

多模态对话。多模态对话主要包含两种任务类型：图像关联对话和图像共享对话。图像关联对话任务以针对问题进行回答（Antol等人，2015；Das等人，2017；Kottur等人，2019）或围绕特定图像创建自然对话（Mostafazadeh等人，2017；Shuster等人，2018；Meng等人，2020；Zheng等人，2021）为核心。相反，图像共享对话任务则专注于选择与提供对话上下文语义上对齐的图像（Zang等人，2021；Feng等人，2022；Lee等人，2023c）。我们采用图像共享对话任务的方法来创建多模态对话，并将这些对话作为图像关联对话任务进行评估。

合成评估基准。 面对人类生成数据的短缺，并观察到大语言模型正接近人类水平的标注质量（Heet al.， 2023； Lee etal.， 2023a），这一发展引发了研究热潮。因此，众多研究开始利用大语言模型来

augment or synthesize large-scale dialogue benchmarks for assessing responses in everyday social interactions (Kim et al., 2023), examining responses in multi-modal environment (Feng et al., 2022), and evaluating responses that align with specific persona (Jandaghi et al., 2023). We leverage LLMs to create data but ensure its high quality with human verification and editing.

3 Generative Pipeline for LoCoMo

An overview of our generative pipeline for LoCoMo is shown in Figure 3. We create two virtual agents, named $\mathcal{L}_1$ and $\mathcal{L}_2$, each initialized with a LLM $\mathcal{M}$ (*i.e.,* `gpt-3.5-turbo`). To start, unique persona statements p are assigned to each agent $\mathcal{L}_i$, ensuring the integration of distinct personalities into their dialogues (§3.1). To mirror real-life experiences, we create a temporal event graph $\mathcal{G}$ for each agent, which illustrates a realistic sequence of life events (§3.2). The LLM agent architecture (Park et al., 2023) is utilized for each agent $\mathcal{L}_i$, enabling them to effectively memorize and reflect conversation history into ongoing dialogues (§3.3). Further, each agent $\mathcal{L}_i$ can share coherent images, thereby enhancing the multi-modal dialogue aspect. Finally, human annotators are tasked with manually filtering and refining the generated data (§3.4).

3.1 Persona

We select an initial persona statement p_c from the MSC dataset (Xu et al., 2022), encompassing 4 to 5 sentences, and employ `gpt-3.5-turbo` as $\mathcal{M}$ to expand these into full persona statement p (See examples and prompt details in Appendix A.1). The generated statements typically include details about one or more of the following elements (Gao et al., 2023a): objectives, past experiences, daily habits, and interpersonal relationships, as well as name, age, and gender of the individual.

3.2 Temporal Event Graph

To utilize the real-life experiences of each agent in the conversation, we construct a temporal event graph, labeled as $\mathcal{G}$, for each agent. This graph $\mathcal{G}$, consisting of events e_i, is produced by applying the condition of $\mathcal{M}$ (i.e., `text-davinci-003`) on a designated persona p. Each event e_i is associated with a date of occurrence t_i. $\mathcal{G}$ includes causal connections $l = (e_i, e_j)$ that illustrate the causal relationships among events $e_i \in \mathcal{G}$ and reflect a natural succession of events in an individual's life. For each $\mathcal{G}$, we create up to 25 events, spread across a time frame of 6 to 12 months, in an iterative process that balances between inference time and the coherence of temporal and causal connections in the timeline. Initially, a small batch of $k = 3$ events is generated, which is then used iteratively as input prompt to create the subsequent batch of k events. See details in Appendix A.2.

3.3 Virtual Agent Architecture

Every agent $\mathcal{L}_i$ incorporates modules from generative agent architecture (Park et al., 2023). The agent has two functions: (1) *reflect & respond*; and (2) *image sharing & image reaction*. The agent is asked to primarily use the *reflect & respond* function while employing *image sharing & image reaction* function judiciously and appropriately within the context of the conversation.

Reflect & Respond. The fundamental process for each agent to *reflect and respond* involves the concept of short-term and long-term memory. During inference, agent $\mathcal{L}_i$ conditions its responses on both short and long-term memories, paralleling how humans remember recent conversations while also recalling distilled important experiences from long-term memory. Following each session k, each agent is asked to produce a summary w_k that is then stored in the short-term $\mathcal{H}_s$. This summary w_k is generated by conditioning $\mathcal{M}$ on both the most recent session conversation history h_k and the preceding summary $w_{k-1} \in \mathcal{H}_l$. For each turn j within session k, a single turn of the conversation h_{k_j} is transformed into an observation o_{k_j} and then stored in the long-term memory $\mathcal{H}_l$. Then, agent $\mathcal{L}_i$ generates a response in session $k + 1$ on the date t^s_{k+1} by basing it on the latest summary w_k, reflections based on the retrieved relevant observations $o \in \mathcal{H}_s$, the ongoing conversation history in the current session h_{k+1} and persona statement p. Long-term temporal narratives are induced in the conversation by additionally conditioning the agent's response on the subset of events in $\mathcal{G}$ that occur between the last and current session i.e. $\{e \in \mathcal{G} \,|\, t^s_k < t^e_i < t^s_{k+1}\}$. See details in Appendix A.2.1.

Image Sharing & Image Reaction. The *image sharing & image reaction* functions are integrated to add a multi-modal dimension to the long-term dialogues.[2] The *image sharing* function is called when the agent decides to send an image. This

[2] Image captions are also saved to long-term memory.

扩充或合成大规模对话基准，用于评估日常社交互动中的响应（Kim et al., 2023），研究多模态环境中的响应（Feng et al., 2022），以及评估与特定角色相符的响应（Jandaghi et al., 2023）。我们利用大语言模型创建数据，但通过人工验证和编辑确保其高质量。

3 LOCOMO生成管道

我们用于 Lo-CoMo 的生成流程概述如图3所示。我们创建了两个虚拟代理，分别命名为 $\mathcal{L}_1$ 和 $\mathcal{L}_2$，每个代理都使用大语言模型 $\mathcal{M}$(即 `gpt-3.5-turbo`) 进行初始化。首先，为每个代理分配独特的角色陈述 p，确保将不同的个性融入他们的对话中 (§3.1)。为了模拟真实生活体验，我们为每个代理创建了一个时间事件图 $\mathcal{G}$，该图展示了一系列真实的生活事件序列 (§3.2)。使用 LLM 代理架构 (Park 等人, 2023) 为每个代理 $\mathcal{L}_i$，使它们能够有效地记忆并将对话历史反映到正在进行的对话中 (§3.3)。此外，每个代理 $\mathcal{L}_i$ 可以分享连贯的图像，从而增强多模态对话方面。最后，人类标注员负责手动过滤和改进生成的数据 (§3.4)。

3.1 角色设定

我们选择一个初始角色陈述 p_c 从MSC数据集（徐等人，2022）中，包含4到5句话，并使用`gpt-3.5-turbo` 作为 $\mathcal{M}$来扩展这些为完整的角色陈述 p (参见附录 A.1中的示例和提示详情)。生成的陈述通常包含以下一个或多个元素（高等人，2023a）的详细信息：目标、过往经历、日常习惯以及人际关系，以及个人的姓名、年龄和性别。

3.2 时间事件图

为了利用每个对话中代理的真实生活经历，我们为每个代理构建一个时间事件图，标记为 $\mathcal{G}$。该图 $\mathcal{G}$ 由事件 e_i 组成，它是通过在指定的角色 p 上应用条件 $\mathcal{M}$ (即 text-davinci-003) 生成的。每个事件 e_i 都与一个发生日期 t_i 相关。 $\mathcal{G}$ 包括因果连接 $l = (e_i, e_j)$，这些连接说明了事件 $e_i \in \mathcal{G}$ 之间的因果关系，并反映了个人生活中事件的自然连续性。对于每个 $\mathcal{G}$，我们创建最多 25 个事件，分布在 6 到 12 个月的时帧内，这是一个平衡推理时间和时间线中时间与因果连接连贯性的迭代过程。最初，会生成一小批 e_i事件，然后将其迭代用作输入提示，以创建后续批次 e_i 的事件。详情见附录A.2。一个时帧内， $k = 3$事件被生成，然后迭代用作输入提示以创建后续批次 k的事件。详情见附录A.2。

3.3 虚拟代理架构

每个代理 $\mathcal{L}_i$ 都包含生成式代理架构的模块（Park等人，2023）。代理有两个功能：(1) 反思与回应；以及 (2) 图像分享与图像反应。代理被要求主要使用 反思与回应 功能，同时在对话的上下文中审慎且恰当地使用 图像分享与图像反应 功能。

反思与回应。每个智能体反思与回应的基本过程涉及短期记忆和长期记忆的概念。在推理过程中，智能体 $\mathcal{L}_i$ 根据短期记忆和长期记忆来生成回应，这类似于人类既记住最近的对话，又从长期记忆中回忆提炼的重要经验。每次会话 k结束后，每个智能体都会被要求生成一个摘要 w_k，然后将其存储在短期记忆 $\mathcal{H}_s$中。这个摘要w_k 是通过结合最新的会话对话历史 h_k 和前一个摘要 $w_{k-1} \in \mathcal{H}_l$来生成的。对于会话 k中的每一轮j 对话，都会将一轮对话 h_{k_j} 转化为一个观察o_{k_j}，然后存储在长期记忆 $\mathcal{H}_l$ 中。然后，智能体 $\mathcal{L}_i$ 在会话 $k + 1$ 的日期 t^s_{k+1} 上生成回应，其依据是最新摘要w_k、基于检索到的相关观察 $o \in \mathcal{H}_s$的反思、当前会话中正在进行的对话历史 h_{k+1} 以及角色声明 p。通过在智能体的回应中额外结合上次会话和当前会话之间发生的部分事件 $\mathcal{G}$（即 $\{e \in \mathcal{G} \,|\, t^s_k < t^e_i < t^s_{k+1}\}$），在对话中诱导出长期时间叙事。详情请参见附录A.2.1。

图片分享&图片反应。 图片分享 & 图片反应 功能被集成，为长期对话增加了多模态维度.[2] 当代理决定发送图片时，会调用<code>图片分享</code>功能。这

[2] 图片说明也会被保存到长期记忆中。

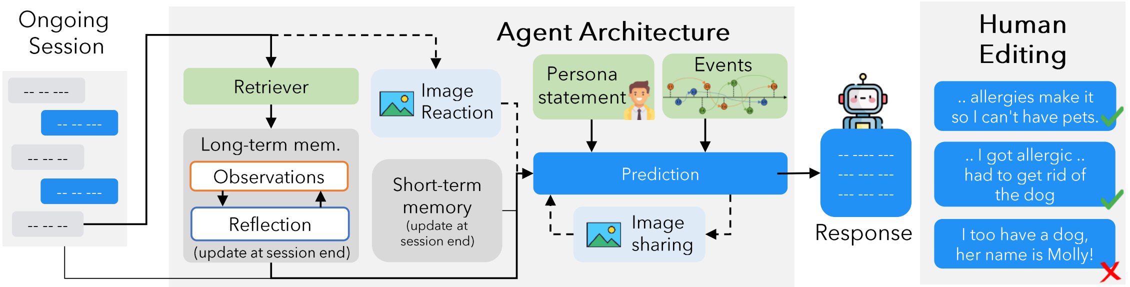

Figure 3: **Overview of the generative pipeline for LoCoMo.** Each LLM agent is assigned a distinct persona and a timeline of causally connected events in their file. The agent is equipped with a memory and reflection module to retrieve relevant history for dialog generation and is also enabled for image-sharing and image-reaction behaviors (left). The generated conversations are edited by human annotators to maintain long-range consistency (right).

process includes: (1) Generate a caption c for the intended image using $\mathcal{M}$; (2) Convert the caption c into relevant keywords w using $\mathcal{M}$; (3) Use the keywords k to find an image through web search $WEB(k)$[3]; (4) Share the chosen $image$. Conversely, the *image reaction* function is triggered upon receiving an image from another agent and entails: (1) Generate caption c for the received image[4]; (2) Generate a reaction for the received image in response using $\mathcal{M}$ (See Appendix A.2.1).

3.4 Human Verification & Editing

In the concluding phase, human annotators are tasked with (1) editing the dialogue to eliminate long-term inconsistencies, (2) removing or substituting irrelevant images, and (3) verifying and editing for alignment between event graphs and the content of the conversations. Overall, we observed that annotators edited nearly 15% of the dialog turns and removed or substituted approx. 19% images present in the LLM-generated dataset. See examples of some edits in Appendix A.3.

4 LoCoMo Evaluation Benchmark

Based on the dialogues generated in section 3, we introduce an evaluation benchmark (see Figure 2) composed of three tasks to assess the accuracy of *long-term memory*. See statistics of the dataset and evaluation benchmark in Table 5 in the Appendix.

4.1 Question Answering Task

A conversational agent is expected to possess a *memory* to remember previous dialogues, reflecting it to create more engaging responses in future conversations. For a comprehensive assessment of this *memory*, we introduce a question-answering task divided into five distinct reasoning categories: (1) **Single-hop** questions require answers based on a single session; (2) **Multi-hop** questions require synthesizing information from multiple different sessions; (3) **Temporal reasoning** questions can be answered through temporal reasoning and capturing time-related data cues within the conversation; (4) **Open-domain knowledge** questions can be answered by integrating a speaker's provided information with external knowledge such as commonsense or world facts; (5) **Adversarial** questions are designed to trick the agent into providing wrong answers, with the expectation that the agent will correctly identify them as unanswerable.

For each category, we calculate the F1 score for exact matches, following the normalization of both the predicted and the actual ground truth answers. However, evaluating long-form answers with automated metrics often presents challenges (Xu et al., 2023). LLMs tend to produce paraphrased responses in varied formats, complicating exact match evaluation. To simplify evaluation in our task, we ensure that answers in our QA annotations are directly taken from the conversations as much as possible. We instruct the LLMs to replicate the exact wording in the conversation when feasible and employ the F1 partial match metric for evaluating the predictions. Each QA sample is also annotated with the turn IDs in the conversation logs that contain the answer. We report the accuracy of retrieving the correct context for RAG models.

4.2 Event Summarization Task

The conversation is generated based on a temporal event graph $\mathcal{G}$ which is constructed by conditioning an LLM on a persona statement p, reflecting the chronological sequence of events in an individual's life. A conversational agent is expected to not only comprehend the causal connections and the

[3]https://pypi.org/project/icrawler/
[4]We use BLIP-2 (Li et al., 2023b) as the captioning model.

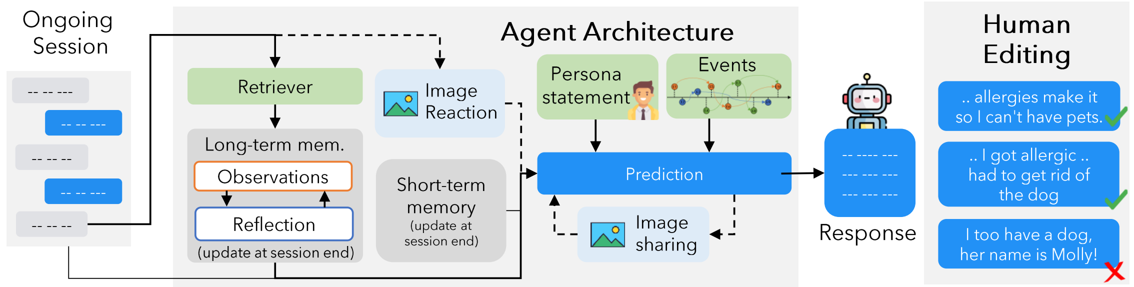

图3：**大语言模型生成LOCOMO的生成流程概述oCoMo**。每个大语言模型代理被分配一个独特的角色和其文件中因果关联的事件时间线。该代理配备记忆和反思模块，用于检索相关历史记录以生成对话，同时支持图像共享和图像反应行为（左侧）。生成的对话由人工标注者编辑以保持长程一致性（右侧）。

该过程包括：(1) 为预期图片生成标题 c 使用 $\mathcal{M}$；(2) 将标题c 转换为相关关键词 w 使用 $\mathcal{M}$；(3) 使用关键词 k 通过网络搜索找到图片$WEB(k)$[3]；(4) 分享选定的 $image$。相反，当从另一个代理接收图片时，会触发<code>图片反应</code>功能，其包括：(1) 为接收到的图片生成标题 c接收到的图片[4]；(2) 使用 $\mathcal{M}$ (参见附录 A.2.1)。

3.4 人工验证与编辑

在收尾阶段，人工标注者负责（1）编辑对话以消除长期不一致，（2）移除或替换无关图像，以及（3）验证和编辑事件图与对话内容之间的对齐。总体而言，我们观察到标注者编辑了约15%的对话回合，并移除或替换了LLM生成数据集中约19%的图像。参见附录A.3中的部分编辑示例

4 LOCOMO评估基准

基于第3节生成的对话，我们引入一个评估基准（见图2），该基准包含三个任务以评估长期记忆的准确性。参见附录中表5 的数据集和评估基准统计

4.1 问答任务

对话代理被期望具备记忆能力，以记住之前的对话，并在未来的对话中将其反映出来，从而创造更吸引人的回复。为了全面评估这种记忆能力，我们引入了一个问答任务该任务分为五个不同的推理类别：(1) **单跳**问题需要基于单个会话回答；(2) **多跳**问题需要综合多个不同会话中的信息；(3) **时序推理**问题可以通过时序推理并捕捉对话中与时间相关的数据线索来回答；(4) **开放域知识**问题可以通过将说话人提供的信息与外部知识（如常识或世界事实）相结合来回答；(5) **对抗**问题旨在诱导代理提供错误答案，期望代理能够正确识别它们为无解问题

对于每个类别，我们计算精确匹配的F1分数，遵循预测答案和实际真实答案的归一化。然而，使用自动指标评估长文本答案通常存在挑战（徐等人，2023）。大语言模型倾向于生成不同格式的释义回答，使精确匹配评估变得复杂。为了简化我们任务的评估，我们尽可能确保QA标注中的答案直接来自对话。我们指示大语言模型在可行时复制对话中的确切措辞，并使用F1部分匹配指标评估预测结果。每个QA样本还标注了包含答案的对话日志中的回合ID。我们报告了RAG模型检索正确上下文的准确率。

4.2 事件摘要任务

该对话基于一个时间事件图 $\mathcal{G}$ 生成，该事件图通过将大语言模型 p 条件化于角色陈述构建而成，反映了个人生活中事件的时间顺序。期望对话代理不仅能够理解因果联系和

[3]https://pypi.org/project/icrawler/ [4]我们使用BLIP-2（李等人，2023b）作为字幕生成模型。

sequence of events in $\mathcal{G}$ but also to recount these events as required. To evaluate the agent's grasp of event dynamics, we introduce the event summarization task which challenges the agent to summarize the events within a designated timeframe and compares the agent's summary with events in $\mathcal{G}$. The events in LoCoMo are densely annotated lists of life events that are hard to summarize due to temporal and causal coreferences present in the dialogues, in contrast to existing summarization benchmarks of research papers (Li et al., 2023a), movie scripts (Chen et al., 2022), books (Kryściński et al., 2022), emails (Zhang et al., 2021b) etc.

Traditional metrics like BLEU (Papineni et al., 2002) and ROGUE (Lin, 2004) focus on lexical similarity between the reference and generated summaries, not meeting our needs as we emphasize factual accuracy in summarization. In this context, we employ FactScore (Min et al., 2023), a method that evaluates the factuality of generated text by decomposing both the reference and hypothesis into atomic facts. We adapt the metric to measure (1) *precision* of the summarized content by counting the number of atomic facts within the content that correspond with those in $\mathcal{G}$; (2) *recall* of the summarized content by determining how comprehensively the atomic facts of $\mathcal{G}$ are represented within the content. We present the F1 score, derived from the calculated precision and recall.

4.3 Multi-Modal Dialogue Generation Task

The conversations in our dataset are anchored to specific personas p and corresponding events $\mathcal{G}$ tailored to p. The topics in conversations evolve from events that were introduced in earlier dialogues, spanning weeks or months. This structure allows for an assessment of whether conversational agents can sustain a coherent persona and a continuous narrative over time. For example, if a speaker recently had an injury, the next conversations would likely focus on them recuperating, rather than engaging in adventurous activities. We assess such consistency by measuring how closely the predicted multi-modal dialogues align with the ground truth multi-modal dialogues in our dataset, quantifying this alignment through MMRelevance (Feng et al., 2022), in addition to other NLG metrics.

5 Experimental Setup

For the question-answering and event summarization tasks, we replace images in LoCoMo with their captions (Li et al., 2023b), and use state-of-art LLMs to reason over text-only dialogues interleaved with image captions. We use images directly for the multimodal dialog generation task only. See additional details in Appendix C.

Question Answering. We evaluate three types of models: (1) **Base** LLMs operating with constrained context lengths where earlier dialogues are omitted i.e., `Mistral-7B-Instruct-v0.2` (Jiang et al., 2023), `Llama-2-70b-chat` (Touvron et al., 2023), and `Llama-3-70B-Instruct`[5]; (2) **Long-context** LLMs with an extended context window i.e., `gpt-3.5-turbo` [6], `gpt-4-turbo` [7], `gemini-1.0-pro` (Team et al., 2023) and `claude-3-sonnet`[8]; (3) **Retrieval-augmented Generation (RAG)** involves retrieving relevant context from a database of dialog history, observations (assertions about speakers; see §3.3, Figure 9), or session-level summaries (see §3.3, Figure 8). We employ DRAGON (Lin et al., 2023) as retriever and `gpt-3.5-turbo` as reader.

Event Summarization. We present experiments using **Base** and **Long-context** setups from the question-answering task, but refrain from including RAG since summarization requires a comprehensive understanding of the entire dialogue, rather than just retrieving a specific portion. We implement incremental summarization i.e., iteratively create a summary of a preceding sessions and then use that summary as a basis to summarize the subsequent sessions (Chang et al., 2023).

Multi-modal Dialogue Generation. We generate 50 conversations using our automated pipeline (without human filtering; §3) for training data and train three versions of MiniGPT-5 (Zheng et al., 2023): (1) **Base** trains on prior dialogue turns only; (2) **+ summary** trains on prior dialogue turns and a global summary of the ongoing conversation; (3) **+ observation** trains on prior dialogue turns and observations retrieved from conversation history. Each run is initialized with a MiniGPT-5 checkpoint finetuned on MMDialog (Feng et al., 2022).

[5]https://llama.meta.com/llama3/
[6]https://platform.openai.com/docs/models/gpt-3-5
[7]https://platform.openai.com/docs/models/gpt-4-and-gpt-4-turbo
[8]https://www.anthropic.com/news/claude-3-family

事件序列在 $\mathcal{G}$，而且还需按要求复述这些事件。为评估代理对事件动态的把握，我们引入了事件摘要任务，该任务要求代理总结指定时间段内的事件，并将代理的摘要与 $\mathcal{G}$ 中的事件进行比较。LoCoMo 中的事件是密集标注的生活事件列表，由于对话中存在时间和因果共指，难以进行总结，这与现有研究论文的摘要基准（李等人，2023a）、电影剧本（陈等人，2022）、书籍（Kry´sci´nski等人，2022）、电子邮件（张等人，2021b）等摘要基准形成对比。

传统的指标如BLEU（Papineni等人，2002）和ROGUE（Lin，2004）侧重于参考文本和生成摘要之间的词汇相似性，这不符合我们的需求，因为我们强调摘要中的事实准确性。在这种情况下，我们采用FactScore（Min等人，2023），这是一种通过将参考文本和假设文本分解为原子事实来评估生成文本事实性的方法。我们将该指标调整为测量 (1)精确率的摘要内容，通过计算内容中与 $\mathcal{G}$中原子事实对应数量的方法；（2）召回率的摘要内容，通过确定 $\mathcal{G}$ 的原子事实在内容中如何全面地被表示。我们提出了基于计算精确率和召回率的F1分数。

4.3 多模态对话生成任务

我们数据集中的对话锚定于特定的角色 p 和相应的事件 $\mathcal{G}$，这些事件针对 p。对话中的主题从先前对话中引入的事件演变，跨越数周或数月。这种结构允许评估对话代理是否能在长时间内保持一致的角色和连续的叙事。例如，如果某个说话人最近受伤，接下来的对话可能会关注他们的康复，而不是参与冒险活动。我们通过测量预测的多模态对话与数据集中真实的多模态对话的吻合程度来评估这种一致性，并通过MMRelevance（冯等人，2022）量化这种吻合度，此外还包括其他NLG指标。

5 实验设置

对于问答和事件摘要任务，我们将LOCOMO中的图像替换为它们的标题（李等人，2023b），并使用最先进的大语言模型对仅包含文本的对话进行推理，这些对话与图像标题交织在一起。我们仅使用图像直接进行多模态对话生成任务。有关更多详细信息，请参阅附录C。

问答。我们评估了三种类型的模型：(1) 在受限上下文长度下运行的基础大语言模型，其中早期对话被省略，即Mistral-7B-Instruct-v0.2（Jiang等人，2023年）、Llama-2-70b-chat（Touvron等人，2023年）和Llama-3-70B-Instruct[5]；(2) 具有扩展上下文窗口的长上下文大语言模型，即gpt-3.5-turbo [6]、gpt-4-turbo 7、gemini-1.0-pro（团队等人，2023年）和claude-3-sonnet[8]；(3) 检索增强生成（RAG）涉及从对话历史数据库、观察（关于说话人的断言；参见§3.3，图9）或会话级摘要（参见§3.3，图8）中检索相关上下文。我们使用DRAGON（Lin等人，2023年）作为检索器，并使用gpt-3.5-turbo作为阅读器。

事件摘要。我们展示了使用 **基础** 和 **长上下文** 设置的问题解答任务实验，但避免包含RAG，因为摘要需要对整个对话进行全面理解，而不仅仅是检索特定部分。我们实现了增量摘要，即迭代创建先前会话的摘要，然后以此摘要为基础来摘要后续会话（Chang 等人，2023）。

多模态对话生成。我们使用自动化流程生成了50个对话（未经人工筛选；§3）作为训练数据，并训练了三个版本的MiniGPT-5（Zheng等人,2023）：(1) **基础**仅在先前的对话回合上训练；(2) + **摘要**在先前的对话回合和当前对话的全局摘要上训练；(3)+ **观察**在先前的对话回合和从对话历史中检索到的观察结果上训练。每次运行都使用在MMDialog（Feng等人, 2022）上微调的MiniGPT-5检查点进行初始化。

[5]https://llama.meta.com/llama3/ [6]
https://platform.openai.com/docs/models/gpt-3 5 [7]-
https://platform.openai.com/docs/models/gpt-4-and-gpt-4-turbo[8]https://www.anthropic.com/news/claude-3-family

Category	Model	Context Length	Answer Prediction (F1)					
			Single Hop	Multi Hop	Temporal	Open Domain	Adversarial	**Overall**
Human	Human	-	95.1	85.8	92.6	75.4	89.4	87.9
Base	`Mistral-7B-Instruct-v0.2`	8K	19.1	15.1	9.3	8.6	28.9	18.7
	`Llama-2-70b-chat`	4K	**20.8**	**18.2**	**15.9**	**18.8**	15.7	18.4
	`Llama-3-70B-Instruct`	4K	17.0	17.0	12.0	13.0	**80.0**	**30.1**
Long context	`gpt-3.5-turbo`	4K	23.8	18.0	15.6	20.4	**34.8**	23.9
		8K	38.5	25.1	22.7	**25.9**	28.7	31.2
		12K	45.7	32.4	**25.5**	23.4	21.5	34.0
		16K	**52.6**	**36.7**	24.3	24.0	14.8	**35.9**
	`gemini-1.0-pro`	1M	62.4	35.3	34.2	19.0	5.2	39.1
	`claude-3-sonnet`	200K	70.7	38.1	26.9	52.2	2.5	42.8
	`gpt-4-turbo`	128K	**72.3**	**51.5**	**51.4**	**38.5**	**15.7**	**51.6**

Table 2: **Question answering performance** of `Base` and `Long-context` models. Optimal performance is in **bold**. Results are based on F1-score for answer prediction; higher is better.

Retrieval Unit	top-k	Answer Prediction (F1 score)						Recall Accuracy (R@k)					
		Single Hop	Multi Hop	Temporal	Open Domain	Adver-sarial	**Overall**	Single Hop	Multi Hop	Temporal	Open Domain	Adver-sarial	**Overall**
`None`	-	29.9	23.3	17.5	29.5	12.8	22.4	-	-	-	-	-	-
`Dialog`	5	53.3	31.2	35.4	**25.0**	21.5	38.8	68.0	35.4	70.4	33.1	43.9	56.7
	10	56.9	34.6	34.5	23.9	**17.5**	39.7	77.7	46.6	77.3	40.8	54.7	66.2
	25	59.9	38.7	**37.2**	25.0	12.8	**41.0**	87.1	62.5	83.5	52.6	66.3	76.7
	50	**60.1**	**40.6**	36.9	22.4	9.9	40.5	**91.1**	**73**	**89.6**	**61.7**	**72.5**	**82.7**
`Observation`	5	54.3	36.3	**40.7**	**26.5**	**32.5**	**43.3**	67.1	41.4	73.1	35.1	37.7	56.2
	10	54.6	39.2	40.5	24.4	28.5	42.8	70.5	50.9	76.4	37.6	45.1	61.3
	25	**54.7**	41.2	38.8	25.8	24.7	42.1	74.7	60.6	80.3	49.0	53.5	67.5
	5	54.1	**41.6**	37.6	24.4	20.4	40.6	**76.3**	**67.8**	**82.0**	**55.9**	**59.5**	**71.2**
`Summary`	2	32.8	22.4	32.9	19.0	**25.3**	29.0	63.0	33.0	57.7	31.5	65.9	65.9
	5	35.1	26.0	37.4	21.2	23.5	30.9	77.0	54.3	72.8	47.6	79.1	72.1
	10	**36.0**	**29.9**	**37.5**	**22.2**	24.0	**32.0**	**88.7**	**72.7**	**84.5**	**67.3**	**88.8**	**84.7**

Table 3: **Question answering performance** of RAG-based `gpt-3.5-turbo`. Optimal performance is in **bold**. Results are based on F1-score metric for answer prediction and recall@k for recall accuracy; higher is better.

6 Experimental Results

We evaluate and analyze the comprehensive performance of all baseline methods for question answering (§6.1), event graph summarization (§6.2), and multi-modal dialogue generation (§6.3).

6.1 Question Answering Task

Tables 2 and 3 present the performance results for the question answering task. We find that: **(1) LLMs with limited context length face challenges in understanding extremely long conversations** due to truncated context windows. Despite `gpt-4-turbo` emerging as the top-performing model with an overall score of 51.6, it notably lags behind the human benchmark of 87.9; **(2) long-context LLMs can comprehend longer narratives, yet they are prone to generating hallucinations**. `gpt-4-turbo` outperforms other approaches on overall performance, but its performance on adversarial questions drops to a mere 15.7%, as compared to 34.8% using `gpt-3.5-turbo` and 80.0% using `llama-3-chat-70B` with 4K context lengths. Similar trends are observed in `gemini-pro-1.5` and `claude-sonnet` models as well. The overall performance of `gpt-3.5-turbo` increases with context length, mainly due to large improvements in single-hop and multi-hop scenarios, yet the performance on adversarial questions drops dramatically. This indicates that LLMs can be easily misled into generating hallucinations when they are subjected to long contexts; (3) **long-context LLMs struggle to utilize the recalled context correctly**. The performance gap between single-hop and multi-hop question categories demonstrates that LLMs have become reasonably good at 'memorizing' a large context window, but find it challenging to perform complex reasoning over the recalled context; **(4) RAG is effective when conversations are stored as observations**. There is a noticeable 5% improvement with `gpt-3.5-turbo` when the input is top 5 relevant observations instead of pure conversation logs. This improvement falters with an increase in the number of retrieved observations, suggesting that it is important to reduce the signal-to-noise (SNR) ratio in retrieved contexts for models to utilize the context accurately. Conversely, using session summaries as context does not significantly improve the performance despite high recall accuracies[9], likely due to loss of information during the conversion of dialogs to summaries.

The interesting finding is that **time reasoning and open-domain knowledge questions are the most challenging scenarios**. (1) LLMs face chal-

[9]For summary-based RAG models, the recall accuracy is based on retrieving the summary of the relevant session(s).

类别	模型	上下文长度	答案预测（F1）					
			单跳	多跳	时间	开放领域	对抗性	**总体**
人类	人类	-	95.1	85.8	92.6	75.4	89.4	87.9
Base	Mistral-7B-Instruct-v0.2	8K	19.1	15.1	9.3	8.6	28.9	18.7
	Llama-2-70b-chat	4K	**20.8**	**18.2**	**15.9**	**18.8**	15.7	18.4
	Llama-3-70B-Instruct	4K	17.0	17.0	12.0	13.0	**80.0**	**30.1**
长上下文	gpt-3.5-turbo	4K	23.8	18.0	15.6	20.4	**34.8**	23.9
		8K	38.5	25.1	22.7	**25.9**	28.7	31.2
		12K	45.7	32.4	**25.5**	23.4	21.5	34.0
		16K	**52.6**	**36.7**	24.3	24.0	14.8	**35.9**
	gemini-1.0-pro	1M	62.4	35.3	34.2	19.0	5.2	39.1
	claude-3-sonnet	200K	70.7	38.1	26.9	52.2	2.5	42.8
	gpt-4-turbo	128K	**72.3**	**51.5**	**51.4**	**38.5**	**15.7**	**51.6**

表2：**问答性能**（基础和长上下文模型）。最佳性能以**粗体**显示。结果基于答案预测的F1分数；越高越好。

检索单元top-k		答案预测（F1分数）						召回准确率（R@k）					
		单 Hop	多 Hop	时间	Open 领域	广告- -字	**总体**	单个 Hop	多 Hop	时间	Open 领域	广告- -srial	**总体**
None	-	29.9	23.3	17.5	29.5	12.8	22.4	-	-	-	-	-	-
对话	5	53.3	31.2	35.4	**25.0**	21.5	38.8	68.0	35.4	70.4	33.1	43.9	56.7
	10	56.9	34.6	34.5	23.9	**17.5**	39.7	77.7	46.6	77.3	40.8	54.7	66.2
	25	59.9	38.7	**37.2**	25.0	12.8	**41.0**	87.1	62.5	83.5	52.6	66.3	76.7
	50	**60.1**	**40.6**	36.9	22.4	9.9	40.5	**91.1**	**73**	**89.6**	**61.7**	**72.5**	**82.7**
观察	5	54.3	36.3	**40.7**	**26.5**	**32.5**	**43.3**	67.1	41.4	73.1	35.1	37.7	56.2
	10	54.6	39.2	40.5	24.4	28.5	42.8	70.5	50.9	76.4	37.6	45.1	61.3
	25	**54.7**	41.2	38.8	25.8	24.7	42.1	74.7	60.6	80.3	49.0	53.5	67.5
	5	54.1	**41.6**	37.6	24.4	20.4	40.6	**76.3**	**67.8**	**82.0**	**55.9**	**59.5**	**71.2**
摘要	2	32.8	22.4	32.9	19.0	**25.3**	29.0	63.0	33.0	57.7	31.5	65.9	65.9
	5	35.1	26.0	37.4	21.2	23.5	30.9	77.0	54.3	72.8	47.6	79.1	72.1
	10	**36.0**	**29.9**	**37.5**	**22.2**	24.0	**32.0**	**88.7**	**72.7**	**84.5**	**67.3**	**88.8**	**84.7**

表3：**基于RAG的gpt-3.5-turbo的问答性能。最佳性能以粗体**显示。结果基于答案预测的F1分数和召回率@k 的召回准确率；越高越好。

6 实验结果

我们评估并分析了所有基线方法在问答（§6.1）、事件图摘要（§6.2）和多模态对话生成（§6.3）方面的综合性能。

6.1 问答任务

表格2 和3 展示了问答任务的性能结果。我们发现：**(1) 上下文长度有限的大语言模型在理解极长对话时面临挑战**由于上下文窗口被截断。尽管gpt-4-turbo 作为表现最佳模型，整体得分为51.6，但明显落后于人 benchmark 的87.9；**(2) 长上下文大语言模型能够理解更长的叙述，但它们容易产生幻觉**。gpt-4-turbo 在整体性能上优于其他方法，但其对抗性问题的表现降至15.7%，而使用gpt-3.5-turbo 时为34.8%，使用llama-3-chat-70B 且上下文长度为4K时为80.0%。在gemini-pro-1.5和claude-sonnet 模型中也观察到类似的趋势。gpt-3.5-turbo 的整体性能随上下文长度增加而提升，这主要归因于单跳和多跳场景的大幅改进，然而其对抗性问题表现急剧下降。这表明当大语言模型面对长上下文时，很容易被误导产生幻觉；(3) **长上下文大语言模型难以正确利用回忆的上下文**。单跳和多跳问题类别的性能差距表明，大语言模型已经相当擅长“记忆”大上下文窗口，但在对回忆的上下文进行复杂推理时仍面临挑战； **(4) 当对话被存储为观察结果时，RAG 是有效的**。当输入为前5个最相关观察结果而非纯对话日志时，gpt-3.5-turbo 的性能提升了5%。随着检索观察结果数量的增加，这种改进会减弱，这表明为了使模型能够准确利用上下文，降低检索上下文中的信噪比（SNR）非常重要。相反，使用会话摘要作为上下文并未显著提升性能，尽管召回率很高[9]，，这可能是由于在将对话转换为摘要的过程中信息丢失了。

有趣的是**时间推理和开放域知识问题是最具挑战性的场景**。(1) 大语言模型面临挑战-

[9]对于基于摘要的RAG模型，召回准确率基于检索相关会话的摘要。

lenges in understanding time concepts within dialogues, which is consistent with findings from other single-turn-based benchmarks focused on temporal reasoning capabilities for LLMs (Wang and Zhao, 2023). (2) LLMs struggle with open-domain knowledge and degrade in the RAG setting. This suggests that while certain open-domain knowledge may be embedded within the model's parameters, introducing improper context from inaccurate retrieval can lead to a decline in performance (Mallen et al., 2023).

6.2 Event Summarization Task

Table 4 presents results for the event summarization task. Powerful long-context models record the highest performance on this task. `gpt-4-turbo` leads to the highest scores in terms of ROUGE and FactScore metrics, followed by `gemini-1.0-pro` and `claude-3-sonnet`. The use of incremental summarization with `Llama-3-70B-Instruct` (4K context window) performs reasonably well compared to the long-context models, demonstrating only a 2.4% drop in ROUGE-L score. However, there is a nearly 10% drop in performance on the FactScore metric, suggesting that it fails to capture as much information as the long-context models. Nonetheless, there remains considerable scope for improving performance on this task. The event summarization task requires long-range dependency to understand the temporal and causal connections between the events discussed by the speaker in multiple sessions (see Figure 7). The wide gap between the best-performing model and the upper bound on this task suggests that **long-context models may not be proficient at utilizing their context appropriately**, which also aligns with similar findings in Li et al. (2023a) as well as the QA task in LoCoMo.

From a manual analysis of predicted summaries, we identify five broad categories of event summarization errors made by LLMs: (1) **missing information** in events because the model fails to make temporal and/or causal connections over a lengthy conversation; (2) **hallucinations** i.e., models pad extra details that are either not present in the conversation or are part of a different event in the same session; (3) errors from **misunderstanding of dialog cues** such as humor or sarcasm is a distinctive issue with comprehension of dialogs; (4) inaccurate **speaker attributions**; and (5) insignificant dialogs that are wrongly considered as **salient** events. See examples in Table 6 in the Appendix.

6.3 Multi-Modal Dialog Generation Task

Figure 4 illustrates the effectiveness of various MiniGPT-5 training variants in multi-modal dialogue generation. Incorporating context into training enhances performance, with the inclusion of observation as context yielding significantly improved results. For instance, in Figure 4A, the retrieved observations contain information about the speaker's experience in video game tournaments, which leads to the prediction of dialog and images that are more faithful to the speaker's persona. This observation is consistent with earlier findings from the QA task as well (see Table 3). Also, we observe that the MM-Relevance score drops with an increase in the length of dialog history (see Figure 4B). Retrieval-augmented generation alleviates the drop in MM-Relevance to some extent.

7 Conclusion

We develop a human-machine pipeline to collect LoCoMo, a dataset of 10 high-quality very long conversations, each encompassing 600 turns and 16K tokens on avg., over up to 32 sessions, and propose an evaluation framework consisting of three tasks that evaluate models' proficiency in long conversations. Our experiments show that LLMs struggle to comprehend long-term narratives within the dialog and fail to draw temporal and causal connections between events discussed by speakers.

8 Limitations

Machine-generated data. Our dataset is sourced primarily from text generated by LLMs. We pursued this method, which has quickly emerged as a popular alternative to time-intensive manual data collection (Kim et al., 2023; Jang et al., 2023b), to avoid the logistical and legal complexities of collecting very long-term real-world conversations at scale. We ensure that the dataset mirrors real-world interactions as much as possible by having human annotators verify and edit the generated conversations. However, we acknowledge that this dataset may not fully reflect the nuances of real-world online conversations.

Limited exploration of multimodal behavior. Since the images in our dataset are sourced from the web, they do not demonstrate the visual long-term consistencies that are usually exhibited in personal photos (e.g., appearance, home environment, people and pets, etc.). Consequently, we find that the

它们在理解对话中时间概念方面存在困难，这与其他专注于大语言模型时间推理能力的单轮基准测试发现一致（王和赵，2023）。(2) 大语言模型在开放域知识方面表现不佳，并在RAG设置中性能下降。这表明虽然某些开放域知识可能嵌入在模型的参数中，但引入来自不准确检索的不当上下文会导致性能下降（Mallen等人，2023）。

6.2 事件摘要任务

表<样式 id='1'>4</样式>展示了事件摘要任务的结果。强大的长上下文模型在这个任务上表现最佳。gpt-4-turbo在ROUGE和事实得分指标上得分最高，其次是gemini-1.0-pro和claude-3-sonnet。使用增量摘要与Llama-3-70B-Instruct（4K上下文窗口）相比长上下文模型表现合理，ROUGE-L得分仅下降2.4%。然而，在事实得分指标上性能下降了近10%，表明它未能捕捉到长上下文模型所包含的更多信息。尽管如此，这个任务仍有相当大的提升空间。事件摘要任务需要长距离依赖关系来理解说话人在多个会话中讨论的事件之间的时间和因果关系（见图<样式 id='3'>7</样式>）。最佳性能模型与任务上限之间的巨大差距表明<样式 id='5'>长上下文模型可能不擅长适当地利用其上下文</样式>，这也与<样式 id='7'>李等人</样式>(<样式 id='9'>2023a</样式>)在LOCOMO的问答任务中的类似发现一致。

从对预测摘要的手动分析中，我们识别出大语言模型在事件摘要中存在的五种主要错误类型：(1) **事件中信息缺失**因为模型未能跨越长时间对话建立时间或因果关系联系；(2) **幻觉**即模型添加了对话中不存在或同一会话中属于不同事件的额外细节；(3) **对话线索理解错误**，例如幽默或讽刺是理解对话时的一个显著问题；(4) **说话人归属不准确**；以及 (5) 被错误地视为 **显著**事件的不重要对话。参见附录中表 6 中的示例。

6.3 多模态对话生成任务

图4展示了各种 MiniGPT-5 训练变体在多模态对话生成中的有效性。将上下文纳入训练可提升性能，其中包含观察作为上下文能显著改善结果。例如，在图 4A 中，检索到的观察信息包含关于说话人在电子竞技比赛中的经历，这导致预测的对话和图像更忠实于说话人的角色。这一观察结果与早期问答任务的发现一致（参见表 3）。此外，我们观察到随着对话历史长度的增加，MM相关性分数会下降（见图 4B）。检索增强生成在一定程度上缓解了MM相关性分数的下降。

7 结论

我们开发了一个人机流程，用于收集L OCOMO数据集，这是一个包含10个高质量超长对话的数据集，每个对话包含600轮对话和平均16K个token，跨越多达32个会话。我们提出了一个包含三个任务的评估框架，用于评估模型在长对话中的能力。我们的实验表明，大语言模型难以理解对话中的长期叙事，并且无法在说话人讨论的事件之间建立时间和因果关系联系。

8 局限性

机器生成数据。 我们的数据集主要来源于大语言模型生成的文本。我们采用了这种方法，它迅速成为了一种流行的替代方案，用于替代耗时的人工数据收集（金等人， 2023； 张等人， 2023b），以避免大规模收集长期真实世界对话的复杂性和法律问题。我们通过让人类标注员验证和编辑生成的对话，确保数据集尽可能反映现实世界的互动。然而，我们承认这个数据集可能无法完全反映现实世界在线对话的细微之处。

对多模态行为的有限探索。 由于我们数据集中的图像来自网络，它们无法展现出个人照片通常具有的视觉长期一致性（例如外貌、家居环境、人和宠物等）。因此，我们发现

Category	Model	Context Length	ROGUE			FactScore		
			ROGUE-1	ROGUE-2	ROGUE-L	Precision	Recall	F1
Base	Mistral-7B-Instruct-v0.2	8K	34.6	10.1	16.4	33.5	31.2	32.3
	Llama-3-70B-Instruct	4K	36.7	11.4	19.2	40.3	35.6	37.8
Long context	gemini-1.0-pro	1M	37.6	13.4	21.1	46.7	42.1	44.2
	claude-3-sonnet	200K	35.1	12.6	21.3	45.6	40.8	43.1
	gpt-4-turbo	128K	**41.2**	**13.8**	**21.6**	**51.9**	**46.5**	**48.9**

Table 4: **Event summarization performance** of Base and Long-context models. The optimal performance is shown in **bold**. Results are based on ROUGE and FactScore (Min et al., 2023) metrics; higher is better.

Figure 4: **Multimodal dialog generation performance of MiniGPT-5**. (A) an example of multimodal dialog predicted using MiniGPT5 with and without observation as retrieved context, (B) Variation of MM-Relevance score with length of dialog history, and (C) comparison of RAG-based MiniGPT-5 methods.

images in our dataset can be replaced with their captions without much loss of information, except for cases where OCR is required. Nevertheless, our work is a first step toward research into the multimodal aspect of very long-term conversations.

Language. Our LLM-based pipeline for generating long-term conversations has been developed for the English language only. However, our pipeline can be made to work with any other language using an LLM that is proficient at that language and appropriate translations of our prompts.

Closed-source LLMs. We use state-of-the-art LLMs in our dialog generation pipeline to create a dialog dataset that is as realistic as possible. Unfortunately, this meant employing LLMs that are not open-sourced and are only available through a paid API, similar to many concurrent works that generate synthetic conversations (Zhong et al., 2023; Lu et al., 2023). We will make the code for our generative pipeline publicly available in the hope that it can be made to work effectively with open-source LLMs in the future.

Evaluation of long-form NLG. LLMs are prone to generating verbose answers even when prompted to answer in short phrases. This creates challenges in evaluating the correctness of answers provided by LLMs and has been widely documented in NLP literature (Chang et al., 2023; Xu et al., 2023; Krishna et al., 2023). Our evaluation framework suffers from the same challenges when used for experimenting with LLMs.

9 Broader Impacts

We adopt and improve a framework of generative agents introduced in Park et al. (2023) for the generation of long-term conversations. Consequently, the ethical concerns of generative agents outlined by Park et al. (2023) apply to our work as well, especially since the goal of our framework is to make the conversations as realistic as possible.

Specifically, conversational agents that can pose as human beings with a realistic life, as enabled by the temporal event graphs in our framework, pose the risk that users may form parasocial relationships with such agents that may affect their lives adversely. We recommend that any practical deployment of the generative framework in our work be always prefaced with a disclaimer about the source of the dialogs.

Second, the use of multimodal LLMs (Zheng et al., 2023) to generate images conditioned on dialog can lead to the propagation of misinformation and social biases, especially if the conversational agent can be coerced into parroting false information or dangerous opinions.

Third, it is tempting to use generative agents to substitute real humans for a process, especially

类别	模型	上下文长度	ROGUE			事实得分		
			ROGUE-1	ROGUE-2	ROGUE-L	精确率	召回率	F1
Base	Mistral-7B-Instruct-v0.2	8K	34.6	10.1	16.4	33.5	31.2	32.3
	Llama-3-70B-Instruct	4K	36.7	11.4	19.2	40.3	35.6	37.8
长上下文	gemini-1.0-pro	1M	37.6	13.4	21.1	46.7	42.1	44.2
	claude-3-sonnet	200K	35.1	12.6	21.3	45.6	40.8	43.1
	gpt-4-turbo	128K	**41.2**	**13.8**	**21.6**	**51.9**	**46.5**	**48.9**

表4：**事件摘要性能**的基础和长上下文模型。最佳性能以**粗体**显示。结果基于ROUGE和事实得分（Min等人，2023）指标；越高越好。

图4：**MiniGPT-5的多模态对话生成性能**。(A) 使用MiniGPT5并使用检索到的上下文作为观察结果预测多模态对话的示例，(B) 对话历史长度与MM相关性分数的变化，(C) RAG-based MiniGPT-5方法的比较。

我们数据集中的图像可以用它们的标题替换，而不会丢失太多信息，除了需要使用OCR的情况。尽管如此，我们的工作是对研究非常长期的对话的多模态方面迈出的第一步。

语言。我们基于大语言模型的长期对话生成管道目前仅支持英语。然而，我们的管道可以通过使用精通该语言的大语言模型和适当的翻译提示来适配任何其他语言。

闭源大语言模型。我们在对话生成管道中使用最先进的大语言模型来创建尽可能真实的对话数据集。不幸的是，这意味着需要使用非开源且仅通过付费API提供的模型，这与许多生成合成对话的并发工作类似（Zhong等人，2023；Lu等人，2023）。我们将公开我们的生成管道代码，希望未来它能有效地与开源大语言模型配合使用。

长文本NLG的评估。大语言模型容易在要求简短回答时生成冗长的答案。这给评估大语言模型提供的答案的正确性带来了挑战，并且已在自然语言处理文献中广泛记录（张等人，2023;徐等人，2023;Krishna等人，2023）。我们的评估框架在用于实验大语言模型时也面临同样的挑战。

9 更广泛的影响

我们采用并改进了Park等人于2023年提出的一个生成式代理框架，用于生成长期对话。因此，Park等人于2023年提出的关于生成式代理的伦理问题同样适用于我们的工作，尤其是由于我们框架的目标是使对话尽可能真实

具体来说，能够像真人一样进行现实生活的对话代理，得益于我们框架中的时间事件图，存在用户可能与此类代理形成准社会关系，从而对他们的生活产生不利影响的风险。我们建议，在我们的工作中，任何生成式框架的实际部署都应始终附上关于对话来源的免责声明。

其次，使用多模态大语言模型Zheng等人，2023，根据对话生成图像，可能导致错误信息的传播和社会偏见，尤其是如果对话代理可以被强迫重复错误信息或危险观点

第三，使用生成式代理来替代人类执行一个流程，尤其是在

when there are significant challenges in working with humans for a particular goal e.g., collecting real-world interactions between humans over a year or more. Care must be taken to ensure that such substitutes are not made in studies whose outcomes may be used to make real-world decisions with tangible impacts on humans. Our work is merely a study of model comprehension in very long-term conversations. We do not make any recommendations for real-world policies based on this study and advise potential users of our framework to avoid making such recommendations as well.

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当在与人类合作以实现特定目标时存在重大挑战时，例如收集一年或更长时间内人类之间的真实世界互动。必须谨慎确保在可能被用于做出对人类产生实际影响的现实世界决策的研究中，不要做出此类替代。我们的工作仅仅是对非常长期对话中模型理解的研究。我们基于这项研究没有对现实世界政策做出任何建议，并建议我们框架的潜在用户也不要做出此类建议。

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Overview

The appendix is organized as follows:
Section A: Details of generative pipeline for the LoCoMo dataset.
Section B: Statistics of LoCoMo dataset, license for data release and annotator details.
Section C: Experimental setup and implementation details.
Section D: Additional results from evaluation on the LoCoMo benchmark.

A Generative Pipeline for LoCoMo

A.1 Persona

We assign unique persona statement p to each agent $\mathcal{L}_i$. For this, we select a range of initial persona statements p_c from the MSC dataset (Xu et al., 2022), each encompassing 4 to 5 sentences. We employ `gpt-3.5-turbo` as $\mathcal{M}$ to expand these into full persona statement p, conditioning $\mathcal{M}$ on the chosen statements p_c. The prompt used for converting a short list of speaker attributes from the MSC dataset (Xu et al., 2022) into a complete persona summary is presented in Fig. 5. We also use a single example of speaker attribute → persona summary as an in-context demonstration along with the prompt. A small selection of personas showcasing the diversity of speakers in the LoCoMo dataset is demonstrated in Fig. 5.

A.2 Temporal Event Graph

As outlined in Sec. 3.2, we use an iterative process for generating event graphs consisting of causally connected events based on a given persona summary. The base prompt for describing the constitution of the event graph, the nature of events and causal connections between events is shown in Fig. 6. First, the base prompt is used along with the prompt for event graph initialization to generate three independent events relevant to a given personality. Then, the base prompt is combined with the prompt for the iterative generation of events to continue generating events that are caused by one or more of the events that are already present in the graph. See an example of a persona and the corresponding temporal event graph in Fig. 7. In the example, Jack aspires to be a hotel manager. Consequently, he enrolls in a hotel management course in July, and after three months, he expresses his excitement about the course on social media. In a similar vein, his passion for gaming results in an invitation from a well-known gaming company.

A.2.1 Virtual Agent Architecture

As outlined in Section 3.3, the virtual agents in our generative pipelines are composed of two mechanisms, *Reflect & respond* (Park et al., 2023) and *Image sharing & response*.

Reflect & respond. This mechanism operates over a combination of short-term and long-term memory. The short-term memory is a summary of a session that is conditioned on the summary from a previous session. See the prompt given to LLMs in our pipeline for generating summaries, and an example of a generated summary, in Fig. 8. The long-term memory is a database of *observations* about each speaker, that are essentially assertive statements about the speaker's persona and life. See the prompt given to LLMs in our pipeline for generating observations, and an example of observations extracted from a conversation, in Fig. 9. In practice, the conversation is annotated with turn IDs for each turn, and the model is also instructed to indicate the turn IDs that directly contribute to each observation. This allows us to keep track of the evidence when using observations as the context for RAG-based models used in our experiments (see Section 5).

Image sharing & response. See prompts for implementing image-sharing and image-response behaviors in Figure 10.

A.3 Human Filtering

Human annotators are instructed to edit the LLM-generated conversations in the following scenarios:

- Remove an image if it is not relevant to the current dialog or the conversation.
- Add context about an image to the current speaker's dialog if it is not discussed by them but the subsequent speaker has reacted to the image.
- Replace an image if it does not match the caption that was used to query for images.
- Edit the dialog when the information present in the dialog is inconsistent with something said (or shared through an image) in earlier or later turns.

概述

附录的结构如下：**A节**：LOCOMO数据集生成流程的详细信息。**B节**：LOCOMO数据集的统计数据、数据发布许可和标注者详情。**C节**：实验设置和实现细节。**D节**：在LOCOMO基准上评估的补充结果。

用于LOCOMO的生成式流程

A.1 角色

我们为每个代理分配独特的角色陈述 p。为此，我们从MSC数据集（徐等人,2022）中选择一系列初始角色陈述 p_c ，每个包含4到5句话。我们使用`gpt-3.5-turbo` 作为 $\mathcal{M}$ 来扩展这些为完整角色陈述 p，以所选陈述为条件 $\mathcal{M}$ 。将MSC数据集中（徐等人， 2022）的说话人属性短列表转换为完整角色摘要的提示显示在图5中。我们还使用一个说话人属性角色摘要的示例作为上下文演示，并附带提示。图5展示了LOCOMO数据集中说话人多样性的少量角色示例。

A.2 时间事件图

如第 3.2节所述，我们使用迭代过程来生成由因果关联事件组成的事件图，这些事件基于给定的角色摘要。描述事件图构成、事件性质以及事件之间因果联系的基础提示如图 6所示。首先，使用基础提示和事件图初始化提示来生成与给定角色相关的三个独立事件。然后，将基础提示与事件迭代生成提示结合，继续生成由图中一个或多个现有事件引起的事件。在图 7 中看到一个角色及其对应的时间事件图的示例。在示例中，杰克渴望成为一名酒店经理。因此，他在7月报名参加了酒店管理课程，三个月后，他在社交媒体上表达了参加课程的兴奋之情。在类似的背景下，他对游戏的热情导致他被一家知名游戏公司邀请。

A.2.1 虚拟代理架构

如第 3.3节所述，我们生成式管道中的虚拟代理由两个机制组成，反思与回应 (Park 等人, 2023)和图像共享与回应。

反思与回应。 该机制在短期记忆和长期记忆的组合上运行。短期记忆是对话会话的摘要，该摘要基于先前会话的摘要进行条件化。参见图8中我们管道中为生成摘要提供给大语言模型的提示，以及一个生成的摘要示例。长期记忆是关于每个说话人的观察数据库，这些观察本质上是对说话人角色和生活的断言性陈述。参见图9中我们管道中为生成观察提供给大语言模型的提示，以及从对话中提取的观察示例。在实践中，对话会为每个回合标注回合ID，模型还会被指示指出直接导致每个观察的回合ID。这使我们能够在使用观察作为我们实验中使用的基于RAG的模型的上下文时跟踪证据（参见第5节）。

图片分享与响应。 请参见图 10中的提示，了解如何实现图片分享和图片响应行为。

A.3 人工过滤

人工标注员被指示在以下场景中编辑大语言模型生成的对话：

- 如果图片与当前对话或整个对话不相关，则移除该图片。
- 如果当前说话人未讨论某张图片，但后续说话人已对该图片做出反应，则向当前说话人的对话中添加关于该图片的背景信息。
- 如果图片与用于查询图片的标题不符，则替换该图片。
- 当对话中的信息与早期或后期回合中所述（或通过图像共享）的内容不一致时，编辑对话。

13864

Prompt

```
Let's write speaker descriptions from a given set of life attributes. Add crucial details in the persona about
the person such as their name, age, marital status, gender, job etc. Add additional details like names of
family/friends or specific activities, likes and dislikes, experiences when appropriate.
```

Examples of generated personas

Ellen is a 55 year old woman who lives in a beautiful Victorian house with her husband. She is a history buff and enjoys collecting antiques and dolls. In her free time, she loves exploring her house's huge attic which holds her collection of old video games. She also loves watching Kpop and taking jazz classes. Her hobbies have allowed her to develop a great sense of rhythm and appreciate all kinds of music. She's also a huge fan of interior design, and loves going out to find additional pieces for her home.

Paul is a 27 year old male who works freelance and lives alone. He enjoys staying active and frequently goes for late night walks. To reward himself for exercising, he often orders pizza, his favorite food. With no one to share it with, he doesn't mind eating several slices at once. He is also a big fan of Dream Theater and loves to listen to their music when out on his walks.

Kylie is a 19 year old female college student in her first year of college. She loves staying active by running, which she does often. She also has a part time job to support her studies. She is single and shares her space with her beloved cat. Kylie has a handful of friends and lively lifestyle.

Figure 5: **Prompt for persona statement (p) generation and examples of personas in LoCoMo.** The prompt used to generate expanded persona statements (p) from initial personas (p_c) for the virtual agents in our conversation generation pipeline (top) and select examples of persona statements present in the LoCoMo dataset.

- Edit the dialog to ensure that the details in the conversation are consistent with those given in the event for the session.
- Remove any events from the event graph if they do not appear in the conversation.

See an example of some edits in Fig. 11.

B Dataset

B.1 Dataset Statistics

See a breakdown of the statistics of the conversations in the LoCoMo dataset in the top panel of Table 5. Also, see a breakdown of the statistics of the annotations in the evaluation benchmark in the bottom panel of Table 5.

B.2 Dataset License

The LoCoMo dataset will be released under the CC BY-NC 4.0 DEED license.[10]

B.3 Annotator Details

The annotators who worked on the LoCoMo dataset were in-house annotators and we were unable to obtain their demographics due to the confidential nature of such information.

Conversation Statistics	# Counts
Total. # conversations h.	10
Avg. # sessions k. in conversation h	27.2
Avg. # turns j. in session k	21.6
Avg. # tokens. conversation h	16,618.1
Avg. # tokens. dialogue h_{k_j} of turn j in session k	29.8
Avg. # tokens. observation o_{k_j} of turn j in session k	19.2
Avg. # tokens. summary w_k of session k	132.4
QA Benchmark Statistics	
# questions. single-hop retrieval	841 (42.3%)
# questions. multi-hop retrieval	282 (14.2%)
# questions. temporal reasoning	321 (16.1%)
# questions. open domain knowledge	96 (4.8%)
# questions. adversarial	446 (22.4%)
Total. # questions.	**1,986**
Event Summarization Statistics	
Avg. # ground truth events. in conversation h	35.8
Avg. # tokens. event summary	1042.7
Multi-modal Dialogue Generation Statistics	
Avg. # images. in conversation h	91.2

Table 5: **Dataset Statistics** of conversation and corresponding benchmark

C Experimental Setup

C.1 Baselines

The conversations in the LoCoMo dataset are composed of natural language dialogs and images that require higher-order reasoning and multimodal coreference resolution, respectively. From initial studies, we observed that multimodal coreference resolution can be performed effectively by replacing images in LoCoMo with their captions gen-

[10] https://creativecommons.org/licenses/by-nc/4.0/

Prompt

让我们根据给定的生活属性集来编写说话人描述。在角色中添加关键细节关于
关于人物的一些细节，例如他们的姓名、年龄、婚姻状况、性别、工作等。添加其他细节，如家庭成员/朋友的名字或特定活动、喜好和厌恶，以及适当时的经历。
家庭成员或朋友，或在适当的时候提及特定的活动、喜好和厌恶。

Examples of generated personas

Ellen is a 55 year old woman who lives in a beautiful Victorian house with her husband. She is a history buff and enjoys collecting antiques and dolls. In her free time, she loves exploring her house's huge attic which holds her collection of old video games. She also loves watching Kpop and taking jazz classes. Her hobbies have allowed her to develop a great sense of rhythm and appreciate all kinds of music. She's also a huge fan of interior design, and loves going out to find additional pieces for her home.

Paul is a 27 year old male who works freelance and lives alone. He enjoys staying active and frequently goes for late night walks. To reward himself for exercising, he often orders pizza, his favorite food. With no one to share it with, he doesn't mind eating several slices at once. He is also a big fan of Dream Theater and loves to listen to their music when out on his walks.

Kylie is a 19 year old female college student in her first year of college. She loves staying active by running, which she does often. She also has a part time job to support her studies. She is single and shares her space with her beloved cat. Kylie has a handful of friends and lively lifestyle.

图 5：**用于生成角色陈述（p）的提示和LOCOMO中的角色示例。** 用于从初始角色（p_c）为我们的对话生成管道中的虚拟代理生成扩展角色陈述（p）的提示（顶部），以及 LOCOMO 数据集中存在的角色陈述的示例。

- 编辑对话以确保对话中的细节与会话事件中提供的信息一致。
- 如果事件不在对话中，则从事件图中删除任何事件。

查看图 11 中的一些编辑示例。。。

B 数据集

B.1 数据集统计

查看 LoCoMo 数据集中对话的统计数据的分解情况，请参阅表 5 的顶部面板。此外，请参阅表 5 的底部面板中评估基准中注释的统计数据分解情况。

B.2 数据集许可

LOCOMO 数据集将根据 CC BY-NC 4.0 DEED 许可发布。[10]

B.3 标注者详情

参与 LOCOMO 数据集标注的标注员是内部标注员，由于此类信息具有保密性质，我们无法获取他们的统计数据。

对话统计	# 计数
总计。# 对话 h。	10
平均。# 会话 k。在对话 h 中	27.2
会话平均回合数 j。k	21.6
对话平均 token 数 h	16,618.1
会话中第 j 回合的对话平均 token 数 h_{k_j}	29.8
会话中第 j 回合的观察平均 token 数 o_{k_j}	19.2
会话第 k 的摘要平均 token 数 w_k	132.4
QA基准统计	
# 问题数。单跳检索	841 (42.3%)
# 问题数。多跳检索	282 (14.2%)
# 问题数。时序推理	321 (16.1%)
# 问题。开放域知识	96 (4.8%)
# 问题。对抗性	446 (22.4%)
总计。# 问题。	**1,986**
事件摘要统计	
对话中平均#真实事件 h	35.8
事件摘要平均#令牌	1042.7
多模态对话生成统计	
对话中平均#图像 h	91.2

表 5：**对话和对应基准**数据集统计

C 实验设置

C.1 基线

LOCOMO数据集中的对话由自然语言对话和图像组成，分别需要更高阶的推理和多模态共指消解。从初步研究来看，我们观察到通过用LOCOMO中的图像替换为其标题，可以有效地进行多模态共指消解。

[10] https://creativecommons.org/licenses/by-nc/4. 0/

Base prompt for event graph generation

```
Let's write a graph representing events that occur in a person's life based on a short summary of their
personality. Nodes represent events and edges represent the influence of past sub-events on a current event.
- The graph is represented in the form of a json list.
- Each entry is a dictionary containing the following keys: "event", “date", "caused_by", "id".
- The "event" field contains a short description of the event.
- The “date" field contains a date.
- The "id" field contains a unique identifier for the event.
- The "caused_by" field represents edges and is a list of "id" of existing events that have caused this event.
Events in the "caused_by" field should occur on dates before the event they have caused. Generate as many
causal connections as possible.
- An example of a causal effect is when the event "started a vegetable garden" causes "harvested tomatoes".
- Events can be positive or negative life events.
```

Additional prompt for event graph initialization

```
For the following input personality, generate three independent events E1, E2 and E3 aligned with their
personality. Events can be positive or negative life events and should reflect evolution in the person's
relationships, state of mind, personality etc.
```

Additional prompt for iterative generation of causal events

```
For the following input personality, generate new events that are caused by one or more EXISTING events. Events
can be positive or negative life events and should reflect evolution in the person's relationships, state of
mind, personality etc. Do not repeat existing sub-events. Start and end your answer with a square bracket.
```

Figure 6: **Prompts for temporal event graph generation.** The prompt used to generate complete personas for the LLMs in our conversation generation pipeline (top) and examples of personas present in the LoCoMo dataset.

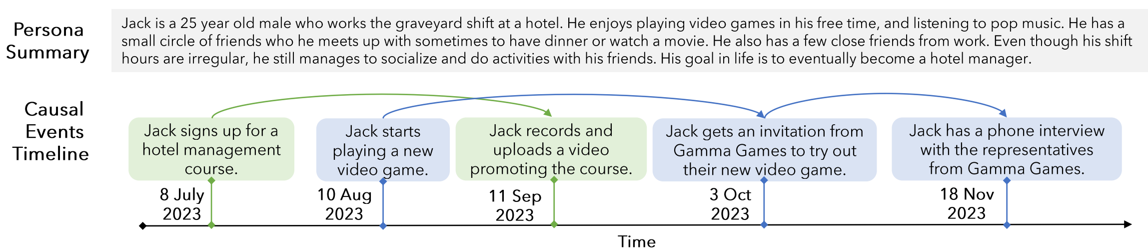

Figure 7: **Temporal Event Graph $\mathcal{G}$ Creation.** Each event is generated in accordance with the specified persona p and causal connections l between events are depicted to illustrate the casual relationships among them.

erated using BLIP-2 (Li et al., 2023b), and using state-of-art LLMs to reason over natural language text interleaved with image captions. Hence, our experiments for the question answering and event summarization tasks are conducted using LLMs. We use the images directly only for experiments on the multimodal dialog generation task.

Question Answering. We carry out experiments using three distinct methodologies: (1) **Base** involves utilizing LLMs to directly conduct the task within a constrained context. The task description comes after the dialogue history. To accommodate the restricted context window size, earlier dialogues are omitted; (2) **Long-context** employs LLMs with an extended context window to expose the models to as much dialogue context as possible; (3) **Retrieval-augmented Generation (RAG)** involves retrieving relevant context from a database of dialog history, observations, or session-level summaries. *Observations* are assertions about each speaker extracted from the dialog history as described in §3.3, see an example in Figure 9. Session-level *summaries* are concise summaries of the conversation that takes place in each session, see an example in Figure 8. For the retrieval model, we employ DRAGON (Lin et al., 2023). To assess the effectiveness in practical scenarios for *Long-context* and *RAG*, we draw comparisons using variants of `gpt-3.5-turbo`. We do not report the performance of long-context fine-tuned open-source models (Chen et al., 2023) or those utilizing sliding window (Bertsch et al., 2023; Dao et al., 2022) due to the variability inherent across different open-source models and the potential reduction in their capability on shorter context.

Event Summarization. We present experiments conducted in two distinct configurations. We use both the **Base** and **Long-context** setups from the

Base prompt for event graph generation

让我们基于对个人性格的简短总结，编写一个表示个人生活中发生的事件的图。节点代表事件，边代表过去子事件对当前事件的影响。- 图以json列表的形式表示。- 每个条目是一个包含以下键的字典："事件"、“日期"、"由...引起"、"id"。- "事件"字段包含事件的简短描述。- “日期"字段包含一个日期。- "id"字段包含事件的唯一标识符。- "由...引起_"字段表示边，是一个现有事件列表的"id"，这些事件导致了当前事件。"由...引起"字段中的事件应在导致它们的事件之前发生日期。- 生成尽可能多的因果连接。- 一个因果效应的例子是事件"开始种蔬菜园"导致"收获西红柿"。- 事件可以是积极或消极的人生事件。

Additional prompt for event graph initialization

对于以下输入的性格，生成三个独立的事件 E1、E2 和 E3，使其与他们的性格相符。事件可以是正面或负面的生活事件，并且应该反映个人在人际关系、心态、性格等方面的演变。

Additional prompt for iterative generation of causal events

对于以下输入的性格，生成由一个或多个现有事件引起的新事件。事件可以是正面或负面的生活事件，并且应该反映个人在人际关系、心态、性格等方面的演变。不要重复现有的子事件。用方括号开始和结束你的答案。

图 6：**用于生成时间事件图的提示。** 用于在我们的对话生成管道中为 LLM 生成完整角色的提示（顶部）以及 LOCOMO 数据集中存在的人物示例。

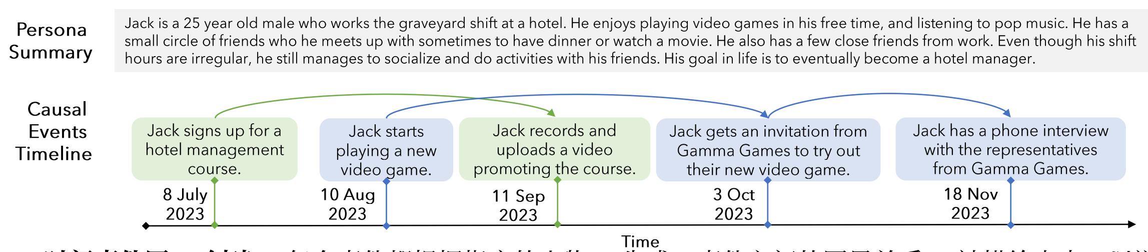

图 7：**时间事件图 $\mathcal{G}$ 创建。** 每个事件都根据指定的人物 p 生成，事件之间的因果关系 l 被描绘出来，以说明它们之间的因果关系。

使用 BLIP-2 (李等人， 2023b)，并使用最先进的 LLM 对自然语言文本与图像标题交替的内容进行推理。因此，我们的问答和事件摘要任务实验使用 LLM 进行。我们仅使用图像直接进行多模态对话生成任务的实验。

问答。 我们采用三种不同的方法进行实验：(1) **基础**方法涉及在受限的上下文中直接使用大语言模型执行任务。任务描述位于对话历史之后。为了适应受限的上下文窗口大小，较早的对话会被省略； (2) **长上下文**方法使用具有扩展上下文窗口的大语言模型，以尽可能多地让模型接触对话上下文； (3)**检索增强生成（RAG）**方法涉及从对话历史、观察结果或会话的数据库中检索相关上下文，会话摘要。 观察结果是从对话历史中提取的关于每个说话人的断言，如 § 3.3所述，图9中有一个示例。会话级摘要是对每个会话中发生的对话的简短总结，图8中有一个示例。对于检索模型，我们采用DRAGON（Lin等人，2023）。为了评估长上下文和RAG在实际场景中的有效性，我们使用`gpt-3.5-turbo`的变体进行比较。我们不报告长上下文微调的开源模型的性能（Chen等人， 2023）或使用滑动窗口（Bertsch等人， 2023； Dao等人,2022）的方法，因为不同开源模型之间存在差异，并且它们在较短的上下文中的能力可能会降低。

事件摘要。 我们展示了在两种不同配置下进行的实验。我们使用了来自**基础**和**长上下文**的设置。

Prompt for generating summaries

In previous interactions, {previous_summary}. The current time and date are {current_date_and_time}. {speaker_1_name} and {speaker_2_name} talked today and had the following conversation: {session} Summarize the interactions between {speaker_1_name} and {speaker_2_name} so far. Include key details about both speakers and include time references wherever possible.

Session (input)

1:14 pm on 25 May, 2023

Hey Caroline, since we last chatted, I've had a lot of things happening to me. I ran a charity race for mental health last Saturday – it was really rewarding. Really made me think about taking care of our minds.

That charity race sounds great, Mel! Making a difference & raising awareness for mental health is super rewarding - I'm really proud of you for taking part!

Thanks, Caroline! The event was really thought-provoking. I'm starting to realize that self-care is really important. It's a journey for me, but when I look after myself, I'm able to better look after my family.

I totally agree, Melanie. Taking care of ourselves is so important - even if it's not always easy. Great that you're prioritizing self-care.

...

...

Previous summary (input)

Caroline and Melanie are old acquaintances who catch up with each other. Caroline attended an LGBTQ support group that she found powerful, particularly inspired by the transgender stories. The support group has made her feel accepted and courageous to embrace herself. She plans to continue her education and explore career options in counseling or mental health to support others with similar issues. Melanie compliments Caroline's empathy and shares a photo of her own painting, highlighting how painting is a great way to express herself and relax. They both agree that taking care of oneself and expressing emotions are important. Melanie mentions that she is going swimming with her kids.

New summary (output)

Caroline and Melanie have been reconnecting as friends. Caroline has been inspired by attending an LGBTQ support group and is now considering a career in counseling or mental health. Melanie, on the other hand, finds her artistic outlet through painting and emphasizes the importance of self-care. Caroline supports Melanie's involvement in a charity race for mental health. They both express the importance of taking care of themselves and their families. Melanie shares her plans for me-time and mentions going swimming with her kids. Caroline reveals her dream of adopting children and shares her research on adoption agencies. Melanie compliments Caroline's kindness and encourages her in the adoption process. Caroline expresses her gratitude for the support she has received and her determination to create a loving home for children.

Figure 8: **Prompt for generating conversation summaries.** The prompt used to iteratively generate a summary for the current session by conditioning on summary from preceding sessions and the raw conversation logs of the current session (top); and an example of inputs for the prompt and corresponding output summary of a session from the LoCoMo dataset.

question answering task, but we refrained from including RAG since summarization requires a comprehensive understanding of the entire dialogue, rather than just retrieving a specific portion. A notable distinction in our approach, compared to the question-answering task, lies in our handling of the context. Specifically, we employ an iterative process of creating a summary of a preceding session and then use that summary as a basis to generate the summary for the subsequent session (Chang et al., 2023). Further, we use a single in-context demonstration of input and output to guide the model toward selecting only significant life events for the summary.

Multi-modal Dialogue Generation. For evaluating multi-modal dialogue generation, we train MiniGPT-5 (Zheng et al., 2023) on 50 conversations generated using our automated pipeline (without human filtering) as detailed in §3. Three distinct versions of the model were developed, each with varying training data: (1) **Base** trains on preceding dialogue turns; (2) **+ summary** trains on both prior dialogue turns and a global summary of the ongoing conversation; (3) **+ observation** trains on both preceding dialogue turns and relevant observations retrieved from the conversation history. For each of these models, we started with a MiniGPT-5 checkpoint pretrained on the MMDialog dataset (Feng et al., 2022).

C.2 Implementation Details

We use OpenAI API, Gemini API, Claude API and Huggingface (Wolf et al., 2020), as of May 2024, with specific settings of $temperature$ set to 0 and top_p set to 1 for evaluation of the LoCoMo benchmark. All experiments, including those for RAG-based models, MiniGPT-5 training, and inference, are conducted on an Nvidia A6000 server with FP32. We report results from a single inference run for each model in our experiments. For MiniGPT-5, we used the hyperparameters recommended in the original codebase and trained our models for 10 epochs, which took approximately 30 hours on a single A6000 GPU.

We use the default implementations of BLEU[11], ROUGE[12], BertScore[13], FactScore[14] metrics in their respective Python packages in our evaluation protocol.

[11] https://www.nltk.org/_modules/nltk/translate/bleu_score.html

[12] https://pypi.org/project/rouge/

[13] https://pypi.org/project/bert-score/

[14] https://github.com/shmsw25/FActScore

Prompt for generating summaries

在之前的交互中，{previous_摘要}。 当前的时间和日期是{current_日期_和_时间。{说话人_1_名称} 和{说话人_2_名称} 今天交谈并进行了以下对话： {会话} 总结一下{说话人_1_名称和{说话人_2_名称迄今为止的交互。包括两位说话人的关键细节，并在可能的情况下包含时间参考。

Session (input)

2023年5月25日下午1:14

嘿，卡罗琳，自从上次聊天以来，我经历了好多事情。我参加了一场为心理健康筹款的慈善赛跑，就在上个周六 – 这件事真的让我受益匪浅。让我深刻意识到要好好照顾自己的心灵关于照顾我们的心灵。

那个慈善赛听起来很棒，梅尔！做出改变 & 为心理健康提高意识是超级有成就感的 - 我为你参加感到非常骄傲！真的很为你感到骄傲！

谢谢，卡罗琳！这个活动真的很有启发性。我开始意识到自我关怀真的很重要。它是一个对我来说，这是一段旅程，但当我照顾好自己时，我就能够更好地照顾我的家人。

我完全同意，梅伦妮。照顾好自己是非常重要的，重要 - 即使它并不总是容易。很高兴你在优先考虑自我关照。

...

...

Previous summary (input)

卡罗琳和梅兰妮是老相识，她们互相联系。卡罗琳参加了一个她觉得很有力量的LGBTQ支持小组，尤其是被跨性别者的故事所鼓舞。支持小组让她感到被接纳，也让她有勇气拥抱自己。她计划继续教育，并探索在咨询或心理健康领域的工作机会，以帮助有类似问题的人。梅兰妮称赞卡罗琳的共情，并分享了自己画作的照片，强调绘画是表达自己和放松的绝佳方式。她们都认为照顾自己和表达情绪很重要。梅兰妮提到她要和孩子们一起去游泳。

New summary (output)

卡罗琳和梅兰妮一直在重新建立友谊。卡罗琳在参加了一个LGBTQ支持小组后受到了鼓舞，现在正在考虑在咨询或心理健康领域发展事业。另一方面，梅兰妮通过绘画找到了自己的艺术出口，并强调了自我关照的重要性。卡罗琳支持梅兰妮参加为心理健康举办的慈善赛跑。她们都表达了照顾自己和家人的重要性。梅兰妮分享了她为自己留出时间的计划，并提到要和孩子们一起去游泳。卡罗琳透露了自己收养孩子的梦想，并分享了关于收养机构的调查。梅兰妮称赞卡罗琳的善良，并鼓励她在收养过程中。卡罗琳表达了对所获得支持的感激之情，以及她为孩子们创造一个充满爱的家的决心。

图8：**用于生成对话摘要的提示**。 该提示用于通过以前会话的摘要和当前会话的原始对话日志作为条件，迭代地生成当前会话的摘要（顶部）；以及LOCOMO数据集的一个会话的提示输入示例和相应的输出摘要。

问答任务，但我们没有包括RAG，因为摘要需要全面理解整个对话，而不仅仅是检索特定部分。与问答任务相比，我们方法的一个显著区别在于我们处理上下文的方式。具体来说，我们采用迭代过程创建前一个会话的摘要，然后使用该摘要作为基础来生成后续会话的摘要（Chang等人,2023）。此外，我们使用一个上下文示例来指导模型仅选择对摘要重要的生活事件。

多模态对话生成。 为了评估多模态对话生成，我们使用自动化流程（无需人工筛选）生成的50个对话，如 § 3所述，训练了MiniGPT-5（Zheng等人， 2023）。我们开发了三个不同版本的模型，每个版本使用不同的训练数据：(1) **基础**在之前的对话回合上训练；(2) **+ 摘要**在之前的对话回合和当前对话的全局摘要上训练；(3) **+ 观察**在之前的对话回合和从对话历史中检索的相关观察结果上训练。对于这些模型中的每一个，我们最初使用一个在MMDi- alog数据集上预训练的MiniGPT-5检查点（冯等人，2022）。

C.2 实现细节

我们使用 OpenAI API、Gemini API、Claude API 和 Huggingface（Wolf 等人，2020），截至 2024 年 5 月，在评估 LOCOMO 基准时，将 $temperature$ 设置为 0，将 top_p 设置为 1。所有实验，包括基于 RAG 的模型、MiniGPT-5 训练和推理，均在配备 FP32 的 Nvidia A6000 服务器上进行。我们报告了每个模型在实验中单次推理运行的结果。对于 MiniGPT-5，我们使用了原始代码库中推荐的超参数，并训练我们的模型 10 个 epoch，在单个 A6000 GPU 上耗时约 30 小时。

我们在评估协议中使用了各自 Python 包中 BLEU[11],ROUGE[12],BertScore[13], FactScore[14] 指标的标准实现。

[11]https://www.nltk.org/_modules/nltk/translate/ 事实得分_.html[12]https://pypi.org/project/rouge/[13] https://pypi.org/project/bert-score/[14] https://github.com/shmsw25/事实得分

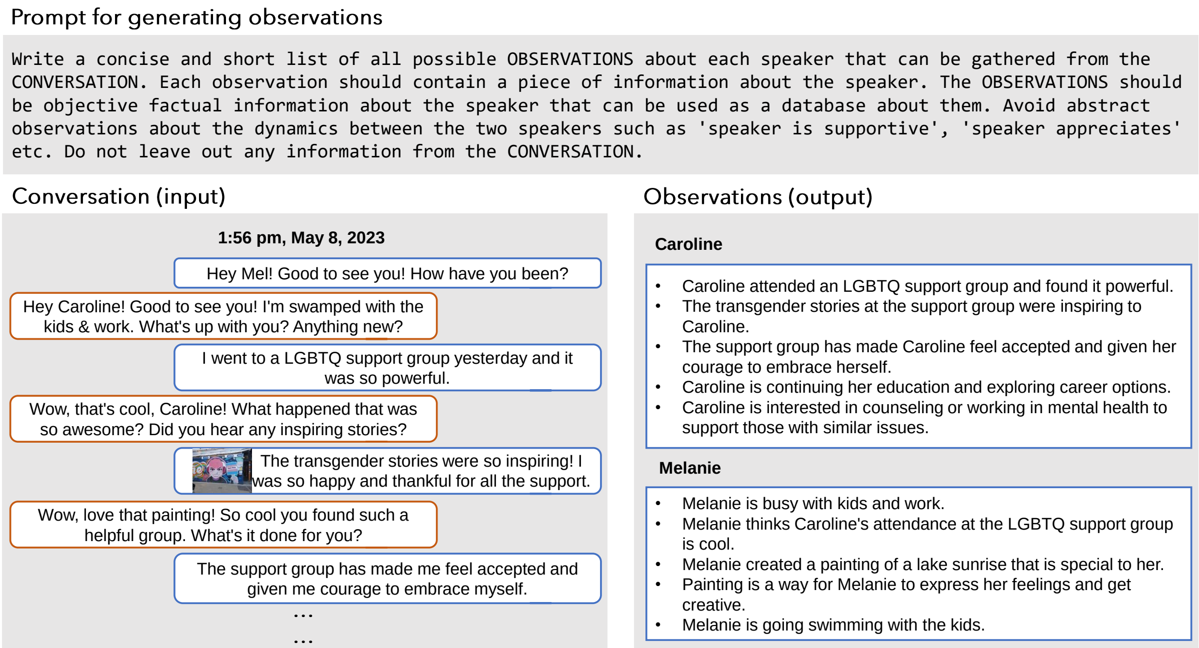

Figure 9: **Prompts for generating observations from conversations.** The prompt used to generate observations from a conversation (top); and an example of inputs for the prompt and corresponding output observations for a session from the LoCoMo dataset.

D Results

D.1 Event Summarization Task

See an example of the five broad categories of event summarization errors made by LLMs, outlined in Section 6.2, in Table 6.

D.2 Multimodal Dialog Generation Task

Results from evaluation of various version of MiniGPT-5 model on the multimodal dialog generation task in the LoCoMo benchmark is shown in Table 7.

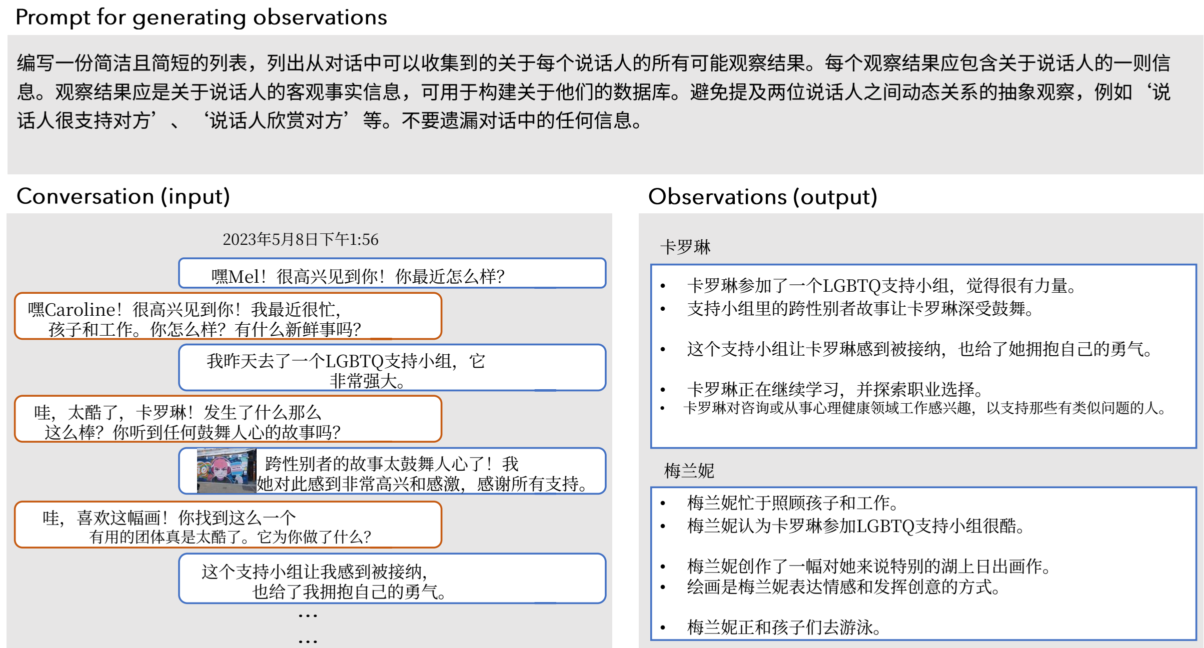

图 9：**从对话中生成观察的提示**。用于从对话中生成观察的提示（顶部）；以及 LOCOMO 数据集中一个会话的输入示例和相应的输出观察。

D 结果

D.1 事件摘要任务

查看 LLM 制作的五类事件摘要错误的示例，这些错误在 6.2 节中概述，见表 6。

D.2 多模态对话生成Task

在 LoCoMo 基准测试中，对 MiniGPT-5 模型不同版本在多模态对话生成任务上的评估结果如表7所示。

Image sharing: Prompt for generated image caption → image query

```
Let's write short image search queries from textual descriptions of photos shared by a user.
Queries should not include names of people, years and other irrelevant details. For example:

Input: That sounds relaxing, Jeremy! As for video game suggestions, have you ever tried "The Legend
of Zelda: Breath of the Wild"? It's an open-world adventure game that I absolutely love. [shares a
photo of Link standing in front of a breathtaking landscape] Have a look at this stunning view!
Output: the legend of zelda: breath of wild link landscape

Input: {generated_image_caption}
Output:
```

Image response: Prompt for shared image caption → response grounded in image

```
{speaker_1_persona}
{speaker_2_persona}

{speaker_1_name} says, {current_turn}, and shares a photo of {shared_image_caption_blip2}. Write
the most natural question or comment {speaker_2_name} can include in their response.
```

Figure 10: **Prompts for image-sharing and image-response behavior.** The prompt used to convert a caption generated by the virtual agent into an image query for the web-based image crawler in our pipeline (top), and the prompt used to generate a response grounded in the image shared by a virtual agent during a conversation as well as the personas of the respective speakers (bottom).

Error Type	Explanation	Ground truth event *or* relevant dialogs	Predicted event
Missing information	Key details about event are omitted because the model fails to make causal and temporal connections over a long conversation.	Joanna submits her third screenplay on loss, identity, and connection to a film contest	Joanna submits her recent screenplay to a film contest.
Hallucination	Non-existent details or details from a different event are padded onto an event	*N*: 'The gaming party was a great success!' *N*: '... said they'd want to do it again next month!' *N*: 'On another note, I made vegan ice cream ...'	Nate's vegan ice cream is a huge success and people want to do it again next month.
Misunder--standing of dialog cues	e.g., model confuses a light-hearted statement from a speaker as a serious statement	*J*: '.. these trails that made me feel like writing a drama.' *N*: '.. go together .. Maybe I'll start to think of a drama myself and write a screenplay ...' *J*: 'Haha, now that would be something! ...'	Nate considers writing his own drama screenplay.
Speaker attribution	Event is attributed to the wrong speaker	Nate invites Joanna to try his homemade lactose-free ice cream.	Joanna invites Nate to her home to try her dairy-free ice cream recipe.
Saliency	Unimportant interactions in the conversation are considered significant by model	*N*: Hey Joanna, what's been up since we last chatted? How's it going?	Nate asks Joanna how she has been she they last talked.

Table 6: **Taxonomy of errors in LLM-generated event summaries.** Five types of errors predominantly occur in the event summaries generated by LLMs. Examples are based on predictions from `gpt-3.5-turbo`.

Category	top-k	BLEU-1/2	Rouge-L	MM-R
Base	-	56.4 / 31.8	11.6	54.2
`+ summary`	1	57.2 / 31.6	11.9	54.7
`+ summary`	2	56.6 / 30.9	11.7	54.1
`+ summary`	5	56.2 / 30.5	11.5	54.0
`+ observation`	5	**58.7 / 32.2**	**12.6**	**55.8**
`+ observation`	10	58.1 / 32.1	12.0	55.1
`+ observation`	25	57.8 / 31.6	11.8	54.9

Table 7: **Multi-modal dialogue generation performance** comparison between different training variants of MiniGPT-5. The optimal performance is shown in **bold**.

Image sharing: Prompt for generated image caption → image query

```
让我们根据用户分享的照片的文本描述来编写简短图像搜索查询。查询不应包含人名、年份和其他无关细节。例如：

输入：听起来很放松，杰里米！至于游戏推荐，你试过《塞尔达传说：荒野之息》吗？它是一款开放世界冒险游戏，我非常
喜欢。[分享了一张林克站在壮丽风景前的照片] 看看这惊人的景色吧！输出：塞尔达传说：荒野之息 林克 风景

输入：{generated_图像_标题输出：
```

Image response: Prompt for shared image caption → response grounded in image

```
{说话人_1_角色}{说话人_2_
角色}

{说话人_1_姓名}说，{当前_回合}，并分享了一张{共享_图像_标题_blip2}。 写出最自然的问题或评论{说话人_2_姓名}可以在他们的回
复中包含。
```

图10：**用于图像分享和图像响应行为的提示**。用于将虚拟代理生成的标题转换为我们的管道中基于网络的图像爬虫的图像查询（顶部），以及用于生成基于虚拟代理在对话期间共享的图像的响应，以及相应说话人角色的提示（底部）。

错误类型	解释	真实事件 或 相关对话	预测事件
缺失信息	关于事件的详细信息被省略，因为模型无法在长时间对话中建立因果关系和时间顺序的联系。	乔安娜提交了她的第三部关于失去、身份和与电影竞赛的联系的剧本。	乔安娜提交了她的最新一部电影竞赛的剧本。
幻觉	不存在的细节或来自不同事件的细节被填充到事件	*N*：'游戏派对非常成功！' *N*：'...他们表示下个月还想再参加！' 内特：'另外，我做了素食冰淇淋……'	内特的素食冰淇淋非常成功，人们想再次尝试，下个月。
误解理解对话提示	例如，模型会将说话人随意的陈述误解为严肃的陈述 将说话人的随意陈述误解为严肃的陈述 陈述	*J*：'..这些让我想写戏剧的痕迹。' *N*：'..可以一起..也许我开始想写戏剧我自己来写一个剧本……' *J*：'哈哈，那可真是了不得！……'	内特考虑写自己的戏剧剧本。
说话人归属	事件被归错给说话人	内特邀请乔安娜尝试他的自制无乳糖冰淇淋奶油。	乔安娜邀请内特去她家尝试她的无乳制品冰淇淋配方。
显著性	对话中不重要互动被模型视为重要	内特：嘿，乔安娜，我们上次聊天以来怎么样了？怎么样？	内特问乔安娜她最近过得怎么样自从他们上次谈话以来。

表 6：**大语言模型生成的事件摘要中的错误分类**。在由大语言模型生成的事件摘要中，主要存在五种类型的错误。示例基于 gpt-3.5-turbo 的预测。

类别	top-k	BLEU-1/2	Rouge-L	MM-R
Base	-	56.4 / 31.8	11.6	54.2
+ 摘要	1	57.2 / 31.6	11.9	54.7
+ 摘要	2	56.6 / 30.9	11.7	54.1
+ 摘要	5	56.2 / 30.5	11.5	54.0
+ 观察	5	**58.7 /32.2**	**12.6**	**55.8**
+ 观察	10	58.1 / 32.1	12.0	55.1
+ 观察	25	57.8 / 31.6	11.8	54.9

表 7：**多模态对话生成性能**在不同 MiniGPT-5 训练变体间的比较。最佳性能以**粗体**显示。

13869

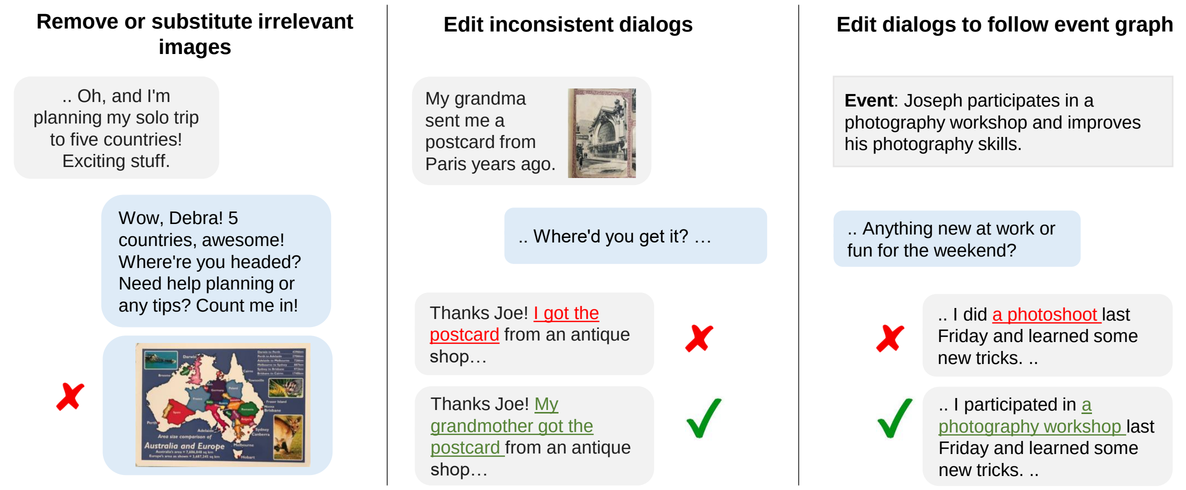

Figure 11: **Example of edits made by annotators.** Human annotators are instructed to make edits in the LLM-generated conversations to remove irrelevant The prompt used to generate complete personas for the LLMs in our conversation generation pipeline (top) and examples of personas present in the LoCoMo dataset.

图11：**标注人员进行的编辑示例**。标注人员被指示对LLM生成的对话进行编辑，以删除无关内容。这是用于我们对话生成管道中为LLMs生成完整角色的提示（顶部），以及LOCOMO数据集中存在的角色示例。

13870